

Boundary Conditions in the B2.5 Code

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March 25, 2002

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1 Introduction

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The boundary conditions are implemented in the B2.5 code by means of the boundary sources, that is in a similar way as the volume sources due to the atomic processes, but they are defined only in the boundary cells of the computational grid. [6] It means that in a case of the boundary conditions at the material surfaces the boundary cells have a role of combined electrostatic sheath and quasineutral collisionless magnetic presheath, which cannot be modelled with the B2.5 fluid equations. According to the existing fluid models of the magnetized boundary plasma, the role of the magnetic presheath, which is located between the collisional plasma presheath and electrostatic sheath, is to deflect the charged particles from their average motion along the magnetic field lines to the direction perpendicular to the wall. The average ion velocity along the magnetic field direction at the entrance of the magnetic presheath from the collisional presheath side is at least the local ion sound velocity, that is $c_{sMPSE} = \sqrt{\frac{k_B(ZT_{eMPSE} + T_{iMPSE})}{m_i}}$ in a two-component plasma, or $u_{\parallel MPSE} \geq c_{sMPSE}$ [10]. The component of the average ion velocity at the entrance of the electrostatic sheath, which is perpendicular to the wall, is also greater than or equal to the local ion sound velocity, i.e., $u_{SE} \geq c_{sSE}$. [4, 10] Since both the magnetic presheath and electrostatic sheath are collisionless regions and assuming that the dissipative effects due to some turbulent mechanisms are negligible in these regions, particle, momentum and energy fluxes are conserved in these regions. The B2.5 equations for the bulk plasma are approximately valid in the collisional presheath, assuming that the charged particle velocity distributions do not differ significantly from Maxwellian or drifting-Maxwellian distributions. (However, it should be emphasized that according to the two-scale kinetic analysis of the boundary plasma near the solid surface, pure Maxwellian or drifting-Maxwellian ion velocity distribution cannot satisfy the generalized Bohm criterion. [4, 12, 17]) It means that they provide the

values for the charged particle densities, drift velocities, ion and electron temperatures at the magnetic presheath edge (*MPSE*), which can be used for calculating the fluxes at the magnetic presheath edge. The value of the poloidal component of the ion drift velocity at the magnetic presheath edge obtained from the bulk transport equations should be corrected, because of the effects related to the stronger local electric field in the magnetic presheath. This electric field causes a strong acceleration of the ions along the magnetic field lines, similar like in a case of a non-magnetic plasma presheath or in a case when the magnetic field is perpendicular to the surface, and additional poloidal $\mathbf{E} \times \mathbf{B}$ drift due to the radial component of the electric field. In some cases $\mathbf{E} \times \mathbf{B}$ drift velocity can be comparable with the local ion sound velocity. [11, 15, 16] The correction of the poloidal component of the ion drift velocity due to the $\mathbf{B} \times \nabla p$ drift is usually negligible, since it contributes mostly to the ion flow parallel to the surface. [8] However, more detailed investigations of the importance of this drift should be performed in the future studies, which should include also kinetic models and numerical simulations (e.g. particle and/or kinetic simulations) of the magnetized boundary plasma.

The boundary sources in the B2.5 code are defined by the boundary cell averages or volume integrals of the negative divergences of the incoming fluxes into the boundary cells. The boundary condition for the continuity equation for each atomic species is given by:

$$S_p = - \int_{B.V.} \nabla \cdot \Gamma \, dV = - \int_{B.S.} \Gamma \cdot d\mathbf{A}, \quad (1)$$

which means that in the discretized form of this equation the value of the particle flux, $\Gamma_j = n \langle v_j \rangle$, at the magnetic presheath edge as a function of the macroscopic plasma parameters should be specified. The boundary condition for the momentum equation is:

$$\mathbf{S}_m = - \int_{B.V.} \nabla \cdot \underline{\mathbf{M}} \, dV = - \int_{B.S.} \underline{\mathbf{M}} \cdot d\mathbf{A}. \quad (2)$$

In this case the appropriate components of the momentum tensor, $M_{jk} = mn \langle v_j v_k \rangle$, at the magnetic presheath edge should be specified in the discretized equations. Similarly, the energy or heat boundary sources are given by:

$$S_q = - \int_{B.V.} \nabla \cdot \mathbf{Q} \, dV = - \int_{B.S.} \mathbf{Q} \cdot d\mathbf{A}, \quad (3)$$

where the appropriate components of the electron and/or ion heat flux, $Q_j = \frac{1}{2}mn \langle v^2 v_j \rangle$, at the magnetic presheath edge as functions of the macroscopic plasma parameters should be specified in the discretized equations. The boundary charge source is given by:

$$S_{ch} = - \int_{B.V.} \nabla \cdot \mathbf{j} dV = - \int_{B.S.} \mathbf{j} \cdot d\mathbf{A}, \quad (4)$$

where the electric current density, $\mathbf{j} = e (\sum_{\alpha} Z_{\alpha} \Gamma_{\alpha} - \Gamma_e)$, at the magnetic presheath edge should be specified in the discretized equations.

2 Analytical Expressions for the Standard (“Old”) Boundary Sources in the B2.5 Code

The boundary sources are expressed as linear and/or weakly non-linear functions of the macroscopic plasma parameters (n_{α} , n_e , $u_{\alpha x}$, $\hat{T}_i$, $\hat{T}_e$, Φ), calculated in the cells neighbouring to the boundary cells. Specific dependences of the boundary sources on these parameters are determined by the choice of the coefficients $C_{pj\alpha}$, $C_{mj\alpha}$, $C_{hij\alpha}$, C_{hej} and C_{chj} (see Section 3). These coefficients are free input parameters, which depend on the theoretical model used for describing the boundary conditions. They must be written by the user in the input file `b2ah.dat`. If `b2ah.dat` file already exists and the boundary conditions for the new simulation are not significantly different, it is practical to write the minor changes of the boundary conditions in the input file `b2mn.dat`. (If the existing `b2ah.dat` file is corrected, it is not necessary to do that.) When the theoretical model predicts more complicated non-linear dependence of the boundary fluxes on the plasma parameters, the effects of these non-linearities can be only approximately taken into consideration in the calculation, e.g., by some appropriate choice of the numerical values of the input parameters (see the example in Section 3). More details about the numerical implementation of the boundary sources can be found in the `b2stbc.f` source code.

2.1 Definitions of Physical Parameters Used in the Analytical Expressions for the Standard Boundary Sources

All following quantities are calculated locally at the appropriate boundary - SOUTH (S), NORTH (N) , WEST (W) and EAST (E) (see Fig. 1, AB

and CD are the divertor plates, x is the poloidal coordinate, y is the radial coordinate, the separatrix goes from W through F and F' to E [6]), i.e., in the boundary cells:

$$S_0 = V/h_{y1}, \quad h_{y1} = h_y \cos(\widehat{xy}) \quad (5)$$

for the SOUTH and NORTH boundary, where V is the volume of the boundary cell, h_y is the y-diameter (radial length) of the boundary cell and $\cos(\widehat{xy})$ is the cosine of the complementary angle between the x-direction and y-direction on the boundary cell;

$$S_0 = V/h_x \quad (6)$$

for the WEST and EAST boundary, where V is the volume of the boundary cell and h_x is the x-diameter (poloidal length) of the boundary cell;

$$S_1 = |S_1^\pm| = \left| \frac{B_y}{B} A \right| \quad (7)$$

for the SOUTH and NORTH boundary, where $S_1^\pm$ is the product of the magnetic field pitch and the surface area on the cell face between the boundary cell and its neighbouring (previous) cell ((ix,-1) and (ix,0) for the SOUTH and (ix,ny) and (ix,ny-1) for the NORTH);

$$S_1 = |S_1^\pm| = \left| \frac{B_x}{B} A \right| \quad (8)$$

for the WEST and EAST boundary, where $S_1^\pm$ is the product of the magnetic field pitch and the surface area on the cell face between the boundary cell and its neighbouring (previous) cell ((-1,iy) and (0,iy) for the WEST and (nx,iy) and (nx-1,iy) for the EAST);

$$c_{s\alpha} = \sqrt{\frac{k_B (Z_\alpha T_e + T_i)}{A_\alpha m_p}} = \sqrt{\frac{k_B (Z_\alpha T_e + T_i)}{m_\alpha}} \quad (9)$$

is the isothermal ion sound velocity for the atomic species α ;

$$c_{st} = \sqrt{\frac{\sum_{\alpha} n_{\alpha} k_B (Z_{\alpha} T_e + T_i)}{\sum_{\alpha} n_{\alpha} A_{\alpha} m_p}} = \sqrt{\frac{\sum_{\alpha} n_{\alpha} k_B (Z_{\alpha} T_e + T_i)}{\sum_{\alpha} n_{\alpha} m_{\alpha}}} \quad (10)$$

is the isothermal ion sound velocity for multispecies plasma;

$$c_{sD\alpha} = \max(c_{s\alpha}, 0.1c_{s\alpha} \pm (B_z/B_x) u_{Dx}) \quad (11)$$

is conditionally corrected ion drift velocity of the atomic species α due to the poloidal $\mathbf{E} \times \mathbf{B}$ drift, where "+" is for the WEST and "-" is for the EAST; the approximate correction is used only if $0.1c_{s\alpha} \pm (B_z/B_x) u_{Dx} > c_{s\alpha}$;

$$c_{stD} = \max(c_{st}, 0.1c_{st} \pm (B_z/B_x) u_{Dx}) \quad (12)$$

is conditionally corrected ion drift velocity of the multispecies plasma due to the poloidal $\mathbf{E} \times \mathbf{B}$ drift, where "+" is for the WEST and "-" is for the EAST; the approximate correction is used only if $0.1c_{st} \pm (B_z/B_x) u_{Dx} > c_{st}$;

$$\rho_{\alpha} = A_{\alpha} m_p n_{\alpha} = m_{\alpha} n_{\alpha} \quad (13)$$

is the mass density of the atomic species α ;

$$n_i = \sum_{\alpha} n_{\alpha} \quad (14)$$

is the total atomic species (ion) particle density;

$$t_{\alpha} = \frac{n_{\alpha}}{n_i} \quad (15)$$

is the relative concentration (particle density) of the atomic species α ;

$$\mathbf{j}_i = \pm e \sum_{\alpha} Z_{\alpha} \Gamma_{\alpha} \quad (16)$$

is the positive ion electric current density in the boundary cells, where "+" is for the NORTH and EAST and "-" is for the SOUTH and WEST;

$$v_{th,e} = \sqrt{\frac{k_B T_e}{m_e}} \quad (17)$$

is the electron thermal velocity;

$$J_e = \frac{1}{\sqrt{2\pi}} e n_e v_{th,e} S_1 \exp\left(-\frac{e\Phi}{k_B T_e}\right) \quad (18)$$

is the electron electric current;

$$\hat{T}_{e,i} = k_B T_{e,i} \quad (19)$$

is the electron/ion temperature in the energy units and

$$\mathbf{Q}_{h\alpha D} = \kappa_{\times\alpha} \left(\mathbf{b} \times \nabla \hat{T}_{\alpha} \right) S_0 \quad (20)$$

is the diamagnetic component of the heat current of the species α , where $\mathbf{b} = \frac{\mathbf{B}}{B}$.

2.2 Standard Boundary Sources

Analytical formulas for the standard boundary sources are presented in this subsection. Their numerical implementation is included in the `b2stbc.f` source code. The B2.5 code architecture is flexible enough, so that it is possible to change the expressions for the boundary sources without changing the other parts of the source code, i.e., only the numerical values of the boundary sources are used in the further calculations.

2.2.1 SOUTH Boundary (Cells (ix,-1))

Particle Sources

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_{\alpha} \quad (21)$$

$$S_{p0\alpha} = C_{p0\alpha}S_0 \quad (22)$$

$$S_{p1\alpha} = C_{p1\alpha}S_0 + C_{p2\alpha}c_{s\alpha}S_0 + C_{p3\alpha}c_{s\alpha}S_1 + C_{p4\alpha}c_{st}S_0 + C_{p5\alpha}c_{st}S_1 \quad (23)$$

Momentum Sources

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_{\alpha x} + S_{m2\alpha}\rho_{\alpha} + S_{m3\alpha}\rho_{\alpha}u_{\alpha x} \quad (24)$$

$$S_{m0\alpha} = C_{m0\alpha}S_0 \quad (25)$$

$$S_{m1\alpha} = 0 \quad (26)$$

$$S_{m2\alpha} = C_{m1\alpha}(-c_{s\alpha})S_1^{\pm} + C_{m2\alpha}(-c_{st})S_1^{\pm} + C_{m5\alpha}\frac{\max(0, \Gamma_{\alpha y})}{n_{\alpha}} \quad (27)$$

$$S_{m3\alpha} = C_{m3\alpha}S_0 + C_{m4\alpha}S_1 + C_{m6\alpha}\frac{\max(0, -\Gamma_{\alpha y})}{n_{\alpha}} \quad (28)$$

All Atom/Ion Heat Source

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i + S_{hi2}n_i + S_{hi3}n_i\hat{T}_i \quad (29)$$

$$S_{hi0} = \sum_{\alpha} [C_{hi0\alpha}t_{\alpha}S_0 + C_{hi6\alpha}\max(0, \Gamma_{\alpha y})] + Q_{hiDy} \quad (30)$$

$$S_{hi1} = \sum_{\alpha} C_{hi1\alpha}t_{\alpha}S_0 + \frac{Q_{hiDy}}{\hat{T}_i} \quad (31)$$

$$S_{hi2} = 0 \quad (32)$$

$$S_{hi3} = \sum_{\alpha} [C_{hi2\alpha}t_{\alpha}c_{s\alpha}S_0 + C_{hi3\alpha}t_{\alpha}c_{s\alpha}S_1 + C_{hi4\alpha}t_{\alpha}c_{st}S_0 + C_{hi5\alpha}t_{\alpha}c_{st}S_1 + C_{hi7\alpha}\frac{\max(0, -\Gamma_{\alpha y})}{n_i}] \quad (33)$$

Electron Heat Source

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e + S_{he2}n_e + S_{he3}n_e\hat{T}_e \quad (34)$$

$$S_{he0} = C_{he0}S_0 + C_{he6}\max(0, \Gamma_{ey}) + Q_{heDy} \quad (35)$$

$$S_{he1} = C_{he1}S_0 + \frac{Q_{heDy}}{\hat{T}_e} \quad (36)$$

$$S_{he2} = 0 \quad (37)$$

$$S_{he3} = C_{he2}v_{th,e}S_0 + C_{he3}v_{th,e}S_1 + C_{he4}c_{st}S_0 + C_{he5}c_{st}S_1 + C_{he7}\frac{\max(0, -\Gamma_{ey})}{n_e} \quad (38)$$

Electric Charge Source

$$S_{ch} = S_{ch0} + S_{ch1}\Phi + S_{ch2}n_e + S_{ch3}n_e\Phi \quad (39)$$

$$S_{ch0} = C_{ch7}J_{ey} - C_{ch6}J_{iy} + \max(C_{ch6}J_{iy}, C_{ch7}J_{ey}) \frac{e\Phi}{\widehat{T}_e} \quad (40)$$

$$S_{ch1} = -\frac{\max(C_{ch6}J_{iy}, C_{ch7}J_{ey})e}{\widehat{T}_e} \quad (41)$$

$$S_{ch2} = C_{ch0}ev_{th,e}S_0 + C_{ch2}ev_{th,e}S_1 + C_{ch4}ec_{st}S_1 \quad (42)$$

$$S_{ch3} = C_{ch1}\frac{e^2v_{th,e}S_0}{\widehat{T}_e} + C_{ch3}\frac{e^2v_{th,e}S_1}{\widehat{T}_e} + C_{ch5}\frac{e^2c_{st}S_1}{\widehat{T}_e} \quad (43)$$

2.2.2 NORTH Boundary (Cells (ix,ny))

Particle Sources

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_\alpha \quad (44)$$

$$S_{p0\alpha} = C_{p0\alpha}S_0 \quad (45)$$

$$S_{p1\alpha} = C_{p1\alpha}S_0 + C_{p2\alpha}c_{s\alpha}S_0 + C_{p3\alpha}c_{s\alpha}S_1 + C_{p4\alpha}c_{st}S_0 + C_{p5\alpha}c_{st}S_1 \quad (46)$$

Momentum Sources

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_{\alpha x} + S_{m2\alpha}\rho_{\alpha} + S_{m3\alpha}\rho_{\alpha}u_{\alpha x} \quad (47)$$

$$S_{m0\alpha} = C_{m0\alpha}S_0 \quad (48)$$

$$S_{m1\alpha} = 0 \quad (49)$$

$$S_{m2\alpha} = C_{m1\alpha}c_{s\alpha}S_1^{\pm} + C_{m2\alpha}c_{st}S_1^{\pm} + C_{m5\alpha}\frac{\max(0, -\Gamma_{\alpha y})}{n_{\alpha}} \quad (50)$$

$$S_{m3\alpha} = C_{m3\alpha}S_0 + C_{m4\alpha}S_1 + C_{m6\alpha}\frac{\max(0, \Gamma_{\alpha y})}{n_{\alpha}} \quad (51)$$

All Atom/Ion Heat Source

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i + S_{hi2}n_i + S_{hi3}n_i\hat{T}_i \quad (52)$$

$$S_{hi0} = \sum_{\alpha} [C_{hi0\alpha}t_{\alpha}S_0 + C_{hi6\alpha}\max(0, -\Gamma_{\alpha y})] - Q_{hiDy} \quad (53)$$

$$S_{hi1} = \sum_{\alpha} C_{hi1\alpha}t_{\alpha}S_0 - \frac{Q_{hiDy}}{\hat{T}_i} \quad (54)$$

$$S_{hi2} = 0 \quad (55)$$

$$S_{hi3} = \sum_{\alpha} [C_{hi2\alpha}t_{\alpha}c_{s\alpha}S_0 + C_{hi3\alpha}t_{\alpha}c_{s\alpha}S_1 + C_{hi4\alpha}t_{\alpha}c_{st}S_0 + C_{hi5\alpha}t_{\alpha}c_{st}S_1 + C_{hi7\alpha}\frac{\max(0, \Gamma_{\alpha y})}{n_i}] \quad (56)$$

Electron Heat Source

$$S_{he} = S_{he0} + S_{he1}\widehat{T}_e + S_{he2}n_e + S_{he3}n_e\widehat{T}_e \quad (57)$$

$$S_{he0} = C_{he0}S_0 + C_{he6}\max(0, -\Gamma_{ey}) - Q_{heDy} \quad (58)$$

$$S_{he1} = C_{he1}S_0 - \frac{Q_{heDy}}{\widehat{T}_e} \quad (59)$$

$$S_{he2} = 0 \quad (60)$$

$$S_{he3} = C_{he2}v_{th,e}S_0 + C_{he3}v_{th,e}S_1 + C_{he4}c_{st}S_0 +$$

$$C_{he5}c_{st}S_1 + C_{he7}\frac{\max(0, \Gamma_{ey})}{n_e} \quad (61)$$

Electric Charge Source

$$S_{ch} = S_{ch0} + S_{ch1}\Phi + S_{ch2}n_e + S_{ch3}n_e\Phi \quad (62)$$

$$S_{ch0} = C_{ch7}J_{ey} - C_{ch6}J_{iy} + \max(C_{ch6}J_{iy}, C_{ch7}J_{ey})\frac{e\Phi}{\widehat{T}_e} \quad (63)$$

$$S_{ch1} = -\frac{\max(C_{ch6}J_{iy}, C_{ch7}J_{ey})e}{\widehat{T}_e} \quad (64)$$

$$S_{ch2} = C_{ch0}ev_{th,e}S_0 + C_{ch2}ev_{th,e}S_1 + C_{ch4}ec_{st}S_1 \quad (65)$$

$$S_{ch3} = C_{ch1}\frac{e^2v_{th,e}S_0}{\widehat{T}_e} + C_{ch3}\frac{e^2v_{th,e}S_1}{\widehat{T}_e} + C_{ch5}\frac{e^2c_{st}S_1}{\widehat{T}_e} \quad (66)$$

2.2.3 WEST Boundary (Cells (-1,iy), Divertor Plate)

Particle Sources

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_{\alpha} \quad (67)$$

$$S_{p0\alpha} = C_{p0\alpha}S_0 + C_{p6\alpha} \max(0, u_{\alpha x}S_1^{\pm}) \quad (68)$$

$$S_{p1\alpha} = C_{p1\alpha}S_0 + C_{p2\alpha}c_{sD\alpha}S_0 + C_{p3\alpha}c_{sD\alpha}S_1 + C_{p4\alpha}c_{stD}S_0 +$$

$$C_{p5\alpha}c_{stD}S_1 + C_{p7\alpha} \max(0, -u_{\alpha x}S_1^{\pm}) \quad (69)$$

Momentum Sources

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_{\alpha x} + S_{m2\alpha}\rho_{\alpha} + S_{m3\alpha}\rho_{\alpha}u_{\alpha x} \quad (70)$$

$$S_{m0\alpha} = C_{m0\alpha}S_0 \quad (71)$$

$$S_{m1\alpha} = 0 \quad (72)$$

$$S_{m2\alpha} = C_{m1\alpha}(-c_{sD\alpha})S_1^{\pm} + C_{m2\alpha}(-c_{stD})S_1^{\pm} + C_{m5\alpha}\frac{\max(0, \Gamma_{\alpha x})}{n_{\alpha}} \quad (73)$$

$$S_{m3\alpha} = C_{m3\alpha}S_0 + C_{m4\alpha}S_1 + C_{m6\alpha}\frac{\max(0, -\Gamma_{\alpha x})}{n_{\alpha}} \quad (74)$$

All Atom/Ion Heat Source

$$S_{hi} = S_{hi0} + S_{hi1}\widehat{T}_i + S_{hi2}n_i + S_{hi3}n_i\widehat{T}_i \quad (75)$$

$$S_{hi0} = \sum_{\alpha} [C_{hi0\alpha}t_{\alpha}S_0 + C_{hi6\alpha}\max(0, \Gamma_{\alpha x})] + Q_{hiDx} \quad (76)$$

$$S_{hi1} = \sum_{\alpha} C_{hi1\alpha}t_{\alpha}S_0 + \frac{Q_{hiDx}}{\widehat{T}_i} \quad (77)$$

$$S_{hi2} = 0 \quad (78)$$

$$S_{hi3} = \sum_{\alpha} [C_{hi2\alpha}t_{\alpha}c_{sD\alpha}S_0 + C_{hi3\alpha}t_{\alpha}c_{sD\alpha}S_1 + C_{hi4\alpha}t_{\alpha}c_{stD}S_0 + C_{hi5\alpha}t_{\alpha}c_{stD}S_1 + C_{hi7\alpha}\frac{\max(0, -\Gamma_{\alpha x})}{n_i}] \quad (79)$$

Electron Heat Source

$$S_{he} = S_{he0} + S_{he1}\widehat{T}_e + S_{he2}n_e + S_{he3}n_e\widehat{T}_e \quad (80)$$

$$S_{he0} = C_{he0}S_0 + C_{he6}\max(0, \Gamma_{ex}) + Q_{heDx} \quad (81)$$

$$S_{he1} = C_{he1}S_0 + \frac{Q_{heDx}}{\widehat{T}_e} \quad (82)$$

$$S_{he2} = 0 \quad (83)$$

$$S_{he3} = C_{he2}v_{th,e}S_0 + C_{he3}v_{th,e}S_1 + C_{he4}c_{stD}S_0 +$$

$$C_{he5}c_{stD}S_1 + C_{he7}\frac{\max(0, -\Gamma_{ex})}{n_e} \quad (84)$$

Electric Charge Source

$$S_{ch} = S_{ch0} + S_{ch1}\Phi + S_{ch2}n_e + S_{ch3}n_e\Phi \quad (85)$$

$$S_{ch0} = C_{ch7}J_{ex} - C_{ch6}J_{ix} + \max(C_{ch6}J_{ix}, C_{ch7}J_{ex}) \frac{e\Phi}{\widehat{T}_e} \quad (86)$$

$$S_{ch1} = -\frac{\max(C_{ch6}J_{ix}, C_{ch7}J_{ex})e}{\widehat{T}_e} \quad (87)$$

$$S_{ch2} = C_{ch0}ev_{th,e}S_0 + C_{ch2}ev_{th,e}S_1 + C_{ch4}ec_{stD}S_1 \quad (88)$$

$$S_{ch3} = C_{ch1}\frac{e^2v_{th,e}S_0}{\widehat{T}_e} + C_{ch3}\frac{e^2v_{th,e}S_1}{\widehat{T}_e} + C_{ch5}\frac{e^2c_{stD}S_1}{\widehat{T}_e} \quad (89)$$

2.2.4 EAST Boundary (Cells (nx,iy), Divertor Plate)

Particle Sources

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_\alpha \quad (90)$$

$$S_{p0\alpha} = C_{p0\alpha}S_0 + C_{p6\alpha}\max(0, -u_{\alpha x}S_1^\pm) \quad (91)$$

$$S_{p1\alpha} = C_{p1\alpha}S_0 + C_{p2\alpha}c_{sD\alpha}S_0 + C_{p3\alpha}c_{sD\alpha}S_1 + C_{p4\alpha}c_{stD}S_0 +$$

$$C_{p5\alpha}c_{stD}S_1 + C_{p7\alpha}\max(0, u_{\alpha x}S_1^\pm) \quad (92)$$

Momentum Sources

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_{\alpha x} + S_{m2\alpha}\rho_{\alpha} + S_{m3\alpha}\rho_{\alpha}u_{\alpha x} \quad (93)$$

$$S_{m0\alpha} = C_{m0\alpha}S_0 \quad (94)$$

$$S_{m1\alpha} = 0 \quad (95)$$

$$S_{m2\alpha} = C_{m1\alpha}c_{sD\alpha}S_1^{\pm} + C_{m2\alpha}c_{stD}S_1^{\pm} + C_{m5\alpha}\frac{\max(0, -\Gamma_{\alpha x})}{n_{\alpha}} \quad (96)$$

$$S_{m3\alpha} = C_{m3\alpha}S_0 + C_{m4\alpha}S_1 + C_{m6\alpha}\frac{\max(0, \Gamma_{\alpha x})}{n_{\alpha}} \quad (97)$$

All Atom/Ion Heat Source

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i + S_{hi2}n_i + S_{hi3}n_i\hat{T}_i \quad (98)$$

$$S_{hi0} = \sum_{\alpha} [C_{hi0\alpha}t_{\alpha}S_0 + C_{hi6\alpha}\max(0, -\Gamma_{\alpha x})] - Q_{hiDx} \quad (99)$$

$$S_{hi1} = \sum_{\alpha} C_{hi1\alpha}t_{\alpha}S_0 - \frac{Q_{hiDx}}{\hat{T}_i} \quad (100)$$

$$S_{hi2} = 0 \quad (101)$$

$$S_{hi3} = \sum_{\alpha} [C_{hi2\alpha}t_{\alpha}c_{sD\alpha}S_0 + C_{hi3\alpha}t_{\alpha}c_{sD\alpha}S_1 + C_{hi4\alpha}t_{\alpha}c_{stD}S_0 + C_{hi5\alpha}t_{\alpha}c_{stD}S_1 + C_{hi7\alpha}\frac{\max(0, \Gamma_{\alpha x})}{n_i}] \quad (102)$$

Electron Heat Source

$$S_{he} = S_{he0} + S_{he1}\widehat{T}_e + S_{he2}n_e + S_{he3}n_e\widehat{T}_e \quad (103)$$

$$S_{he0} = C_{he0}S_0 + C_{he6}\max(0, -\Gamma_{ex}) - Q_{heDx} \quad (104)$$

$$S_{he1} = C_{he1}S_0 - \frac{Q_{heDx}}{\widehat{T}_e} \quad (105)$$

$$S_{he2} = 0 \quad (106)$$

$$S_{he3} = C_{he2}v_{th,e}S_0 + C_{he3}v_{th,e}S_1 + C_{he4}c_{stD}S_0 +$$

$$C_{he5}c_{stD}S_1 + C_{he7}\frac{\max(0, \Gamma_{ex})}{n_e} \quad (107)$$

Electric Charge Source

$$S_{ch} = S_{ch0} + S_{ch1}\Phi + S_{ch2}n_e + S_{ch3}n_e\Phi \quad (108)$$

$$S_{ch0} = C_{ch7}J_{ex} - C_{ch6}J_{ix} + \max(C_{ch6}J_{ix}, C_{ch7}J_{ex})\frac{e\Phi}{\widehat{T}_e} \quad (109)$$

$$S_{ch1} = -\frac{\max(C_{ch6}J_{ix}, C_{ch7}J_{ex})e}{\widehat{T}_e} \quad (110)$$

$$S_{ch2} = C_{ch0}ev_{th,e}S_0 + C_{ch2}ev_{th,e}S_1 + C_{ch4}ec_{stD}S_1 \quad (111)$$

$$S_{ch3} = C_{ch1}\frac{e^2v_{th,e}S_0}{\widehat{T}_e} + C_{ch3}\frac{e^2v_{th,e}S_1}{\widehat{T}_e} + C_{ch5}\frac{e^2c_{stD}S_1}{\widehat{T}_e} \quad (112)$$

Figure 1: Schematic view of the computational mesh.

3 Standard Boundary Sources near the Material Surfaces (Divertor Plates)

3.1 Introduction

The choice of the free input parameters, which determine the standard boundary sources, will be presented on the example of a two-component, non-drifting ($u_{\parallel\infty} = 0$) plasma (positive ions and electrons) near the divertor plates. Ion reflection and emission from the surface are neglected. [4, 14] In the B2.5 code, these boundary sources are the boundary conditions at the WEST and EAST boundary (see Fig. 1). Two cases of electric current collection by the divertor plates will be considered:

- electrically floating divertor plates, which means that the net electric current collected by the plates is zero, i.e., $I_d = 0$, and
- electrically biased divertor plates, when the net electric current collected by the plates is non-zero, i.e., $I_d \neq 0$.

In the second case it is assumed that the net current does not cause some macroscopic current-driven plasma instabilities and that the spatial variation of the electric potential near the wall is qualitatively similar as in the floating case.

3.2 Standard Particle Sources

The poloidal component of the ion flux at the magnetic presheath edge (*MPSE*) is given by:

$$\Gamma_{xMPSE} = n_{MPSE} \langle v_x \rangle_{MPSE}, \quad (113)$$

where n_{MPSE} is the ion particle density and $\langle v_x \rangle_{MPSE}$ is the poloidal component of the ion drift velocity, both calculated at the magnetic presheath edge. According to the fluid model [11], equation (113) can be rewritten as:

$$\Gamma_{xMPSE} = n_{MPSE} u_{\parallel MPSE} \sin \theta, \quad (114)$$

where $\sin \theta = \frac{B_x}{B}$ and

$$u_{\parallel MPSE} = c_{sMPSE} \pm \mathcal{O} \left(\frac{u_{DMPSE}}{\tan \theta} \right), \quad (115)$$

where $c_{sMPSE} = \sqrt{\frac{k_B(T_{eMPSE} + T_{iMPSE})}{m_i}}$ is the local ion sound velocity and $u_{DMPSE} = \left(\frac{E_y}{B} \right)_{MPSE}$ is the local drift velocity due to the poloidal $\mathbf{E} \times \mathbf{B}$ drift in the magnetic presheath (in the B2.5 geometry this drift is caused by the radial electric field E_y). Equations (11) and (12) for conditionally corrected ion drift velocities $c_{sD\alpha}$ and c_{stD} (the numerical factor 0.1 is due to a specific value of $\sin \theta$, for which the value of the ion drift velocity should be corrected) are based on the approximation:

$$u_{\parallel MPSE} = c_{sMPSE} \pm \frac{u_{DMPSE}}{\tan \theta}. \quad (116)$$

Better approximation and more complicated analytical formula for $u_{\parallel MPSE}$ can be found in the article of P. C. Stangeby and A. V. Chankin [11] (the

coordinate system in their calculation is different, x is the radial component and y is the poloidal component). According to their model, the linear approximation of the expression for $u_{\parallel MPSE}$ is:

$$u_{\parallel MPSE} = c_{sMPSE} - \frac{l_{sh}}{2|\Psi_W|\tan\theta} \left(\frac{1}{l_v} - \frac{1}{l_n} \right) u_{DMPSE}, \quad (117)$$

where Ψ_W is normalized sheath electric potential fall (typically $|\Psi_W| \approx 3$ for hydrogenic plasmas) and l_{sh} , l_v and l_n are characteristic decay lengths for the sheath electric potential, ion drift velocity and particle density respectively. [11] Assumption $l_v = l_n$ in equation (117) gives zero contribution of the electric drift term, which is probably physically unrealistic, because it means that even very large poloidal $\mathbf{E} \times \mathbf{B}$ drifts, so that local $c_s \approx \frac{E_y}{B}$, do not contribute in the linear order to the ion drift velocity at the magnetic presheath edge. Implementation of their non-linear expression for $u_{\parallel MPSE}$ into the B2.5 code demands significant changes of the standard source code.

The alternative approach is to specify the ion flux at the electrostatic sheath edge:

$$\Gamma_{xSE} = n_{SE} \langle v_x \rangle_{SE}, \quad (118)$$

which is equal to the ion flux at the magnetic presheath edge Γ_{xMPSE} . In this case, it can be assumed that:

$$\langle v_x \rangle_{SE} = c_{sSE} \sin\theta. \quad (119)$$

However, the ion density at the electrostatic sheath edge n_{SE} is different from the ion density at the magnetic presheath edge n_{MPSE} and it can be some very complicated non-linear function of other plasma parameters. In order to apply the expressions for the particle, momentum and energy fluxes at the electrostatic sheath edge, it is necessary to develop a kinetic model of the collisionless plasma region near the solid surface, because the fluid models cannot describe correctly this region. In principle, the validity of the B2.5 equations is limited only to the plasma regions where the ion velocity distributions do not differ significantly from Maxwellian or drifting-Maxwellian distributions, which is not the case also in the plasma presheath. The correct two-scale or multi-scale analysis of the boundary plasma near the divertor plate should include a separate description of the collisional

presheath, which is located between the bulk plasma and collisionless magnetic presheath. [4] Another possibility is to develop a model of a complete plasma presheath, which includes the collisional presheath and collisionless magnetic presheath. In any case, the solutions of the equations for these different plasma regions should be smoothly connected and implemented into the boundary conditions of the B2.5 code.

In the standard version of the boundary sources, the particle source can be specified by choosing $C_{p3} = 1$ and all other $C_{pj} = 0$. The numerical factor 0.1 in the expressions for $c_{sD\alpha}$ and c_{stD} (see equations (11) and (12)) can be replaced by $\sin\theta$, so that they can be more general. Another possibility is to redefine $c_{sD\alpha}$ given by equation (11), so that the linearized expression for the particle flux of species α will be:

$$\Gamma_{\alpha x MPSE} = n_{\alpha MPSE} (c_{s\alpha MPSE} \sin\theta \pm a_D u_{D MPSE} \cos\theta), \quad (120)$$

where the additional numerical factor a_D is calculated with some theoretical model of the magnetic presheath (e.g., [11]). In the absence of any reliable theoretical model it can be assumed that $a_D = 0$ or $a_D = 1$, which corresponds to the electric drift-free case and to the approximation (116) respectively. If these small changes of the source code are applied, the appropriate choice of the free input parameters for the particle sources will be again $C_{p3\alpha} = 1$ and all other $C_{pj\alpha} = 0$.

However, many experiments (see [3], [5], [9], [13]) and some numerical simulations [1] have shown that when the angle θ is very small, i.e., especially for $\theta \sim \sqrt{\frac{m_e}{m_i}}$, which is in practical cases usually for $\theta \sim 5^\circ$, the expression for the ion flux (114) is not correct. In such cases, the measured charged particle fluxes to the surface are anomalously high compared with the theoretical prediction given by equation (114). They can be approximately described with a turbulent diffusion model:

$$\Gamma_{x MPSE} = -D_{turb} \partial_x n_{MPSE} \approx \frac{D_{turb}}{\lambda_{MPSx}} n_{MPSE}, \quad (121)$$

where D_{turb} is the local turbulent diffusion coefficient and λ_{MPSx} is the characteristic poloidal dimension of the magnetic presheath. The quotient $\frac{D_{turb}}{\lambda_{MPSx}}$ has a dimension of a velocity and the approximation of the right-hand side of equation (121) enables its implementation into the B2.5 code by using

the expressions for the boundary sources. In principle, only the theoretical model of turbulent plasma transport for this specific case can give a correct result for the coefficient S_{p1} (see equations (69) and (92)). In the absence of such a theoretical model it is necessary to estimate the quotient $\frac{D_{turb}}{\lambda_{MPsx}}$ properly, so that the numerical values of the free input parameters for the particle sources can be chosen.

According to the fluid models of the magnetized boundary plasma, the characteristic length scale of the magnetic presheath is $\rho_s \cos \theta$, where $\rho_s = \frac{c_s}{\omega_i}$ is the Larmor radius for the ions moving with the ion sound velocity. Since in our case $\theta \sim \sqrt{\frac{m_e}{m_i}} \ll 1$, λ_{MPsx} is of the order of magnitude of ρ_{sMPSE} , i.e., $\lambda_{MPsx} \sim \rho_{sMPSE}$. The turbulent diffusion coefficient D_{turb} is approximately given by:

$$D_{turb} \approx \sqrt{\langle \tilde{u}_D^2 \rangle} \lambda_c, \quad (122)$$

where $\tilde{u}_D = \frac{\tilde{E}_y}{B}$ is the drift velocity due to the fluctuating (turbulent) electric field $\tilde{E}_y$ and $\lambda_c \approx \langle k_y \rangle^{-1}$ is the characteristic coherence length of the turbulent spectrum. Turbulent fluctuations of the magnetic field are much smaller and can be neglected in the study of the particle transport. (However, they can be important for the heat transport. [7]) Assuming $\sqrt{\langle \tilde{E}_y^2 \rangle} \approx \sqrt{\langle \tilde{\Phi}^2 \rangle} \langle k_y \rangle \equiv \delta \Phi \langle k_y \rangle$, equation (122) can be rewritten as:

$$D_{turb} \approx \frac{\delta \Phi}{B}. \quad (123)$$

Turbulent fluctuations of the electric potential $\delta \Phi$ are approximately proportional to the turbulent fluctuations of the plasma density, i.e.,

$$\frac{\delta n}{\langle n \rangle} \approx \frac{e \delta \Phi}{k_B T_e}, \quad (124)$$

where so-called Boltzmann's fluctuations are assumed and possible decorrelations between the potential and density fluctuations are neglected. In the electrostatic turbulence, the fluctuations of the electron temperature are usually much smaller than the density and potential fluctuations, so that they can be also neglected. Equations (123) and (124) can be combined to give:

$$D_{turb} \approx \frac{\delta n}{\langle n \rangle} \frac{k_B T_e}{eB}. \quad (125)$$

This simple calculation gives the characteristic Bohm $\frac{1}{B}$ -scaling of the anomalous diffusion coefficient. In the original Bohm diffusion coefficient, there is an empirical numerical factor $\frac{1}{16}$ instead of $\frac{\delta n}{\langle n \rangle}$. The physical picture of the result (125) is clear: increased turbulence level results in increased turbulent transport of the charged particles. Assuming $\lambda_{MPSE} \sim \rho_{sMPSE}$, the quotient $\frac{D_{turb}}{\lambda_{MPSE}}$ is approximately:

$$\frac{D_{turb}}{\lambda_{MPSE}} \approx \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE} c_{sMPSE}. \quad (126)$$

This result qualitatively explains the anomalously high particle fluxes at small grazing angles θ . Despite the fact that collisions in the divertor plasma can significantly reduce the turbulent fluctuation level, it is still valid that $\frac{\delta n}{\langle n \rangle} \gg \sin \theta$ in such cases. If the information about the density fluctuation level $\frac{\delta n}{\langle n \rangle}$ is available, e.g., from the experiment, this simple result can be used in the expression for the particle source by choosing $C_{p2} = \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE}$ and all other $C_{pj} = 0$. The other possibility is to calculate $\frac{D_{turb}}{\lambda_{MPSE}}$ somehow and choose $C_{p1} = \frac{D_{turb}}{\lambda_{MPSE}}$ and all other $C_{pj} = 0$. It is also important to mention that the turbulent transport near the material wall can be different from the turbulent transport in the bulk plasma, which means that the corresponding macroscopic turbulent transport coefficients can be different.

Equations (114) and (121) may be considered as limiting cases, because in the general case or for some values of the grazing angle θ the classical and turbulent contributions to the plasma transport can be comparable by magnitude. In the bulk plasma transport there are also contributions due to the classical and turbulent transport. Thus it is reasonable to assume the following expression for the ion flux at the magnetic presheath edge:

$$\Gamma_{xMPSE} = n_{MPSE} (c_{sMPSE} \sin \theta \pm a_D u_{DMPSE} \cos \theta) - D_{turb} \partial_x n_{MPSE}. \quad (127)$$

Equation (127) cannot be directly implemented in the standard expressions for the boundary sources, because they do not include derivatives of the

plasma parameters, such as $\partial_x n_\alpha$, $\partial_x \widehat{T}_i$, $\partial_x \widehat{T}_e$ etc. There are two practical solutions for this problem. The first solution is to add in the source code new terms with the derivatives of the plasma parameters in the expressions for the boundary sources, so that equation (127) can be implemented directly. The second solution is to estimate the turbulent component of the ion flux, as it was shown in the case of a very small grazing angle θ , so that the appropriate choices of the free input parameters will be $C_{p1} = \frac{D_{turb}}{\lambda_{MPSE}}$, $C_{p3} = 1$ and all other $C_{pj} = 0$, or $C_{p2} = \left(\frac{\delta n}{\langle n \rangle}\right)_{MPSE}$, $C_{p3} = 1$ and all other $C_{pj} = 0$.

3.3 Standard Momentum Sources

In order to calculate the boundary sources for the momentum conservation equation, for $d\mathbf{A} = dA\mathbf{e}_x$ equation (2) can be rewritten as:

$$\mathbf{S}_m = - \int_{B.S.} \underline{\mathbf{M}} \cdot \mathbf{e}_x dA \equiv - \int_{B.S.} \mathbf{P}_m dA, \quad (128)$$

where the momentum flux $\mathbf{P}_m$ is given by [2]:

$$\mathbf{P}_m = m_i n \left[\langle v_x^2 \rangle, \langle v_y v_x \rangle, \langle v_z v_x \rangle \right]^T. \quad (129)$$

All quantities in equation (129) should be calculated at the magnetic presheath edge (*MPSE*). The velocity $\mathbf{v}$ may be decomposed into an average part $\mathbf{u}$ and a random part $\mathbf{w}$, associated with a “thermal” motion, i.e.,

$$\mathbf{v} = \mathbf{u} + \mathbf{w}, \quad (130)$$

where $\langle \mathbf{w} \rangle = 0$. Then the components of the momentum flux $\mathbf{P}_m$ can be written as:

$$P_{mx} = m_i n \left(u_x^2 + \langle w_x^2 \rangle \right), \quad (131)$$

$$P_{my} = m_i n \left(u_y u_x + \langle w_y w_x \rangle \right), \quad (132)$$

$$P_{mz} = m_i n (u_z u_x + \langle w_z w_x \rangle). \quad (133)$$

In further derivations some assumptions about the ion velocity distribution at the magnetic presheath edge should be made. The assumption that the ion velocity distribution is drifting-Maxwellian is acceptable from the point of view of the bulk plasma transport, but from the point of view of the two-scale analysis of the plasma-sheath problem such an ion velocity distribution cannot satisfy the generalized Bohm criterion [4]. In the absence of any better approximation and assuming that $u_z \approx 0$ and $u_y \approx 0$, the only non-zero component of $\mathbf{P}_m$ is P_{mx} , because for the Maxwellian distribution the anisotropic shear (viscosity) terms $m_i n \langle w_y w_x \rangle$ and $m_i n \langle w_z w_x \rangle$ are zero. This assumption is maybe acceptable for the component P_{mz} , but it may be very questionable for the component P_{my} , because, according to the study of A. V. Chankin and P. C. Stangeby [8], the radial component of the diamagnetic drift velocity:

$$\mathbf{u}_{Dm} = \frac{\mathbf{B} \times \nabla p}{enB^2}, \quad (134)$$

which is in our case:

$$u_{Dmy} = \frac{B_z \partial_x p - B_x \partial_z p}{enB^2}, \quad (135)$$

and the corresponding particle flux $\Gamma_{Dmy} = nu_{Dmy}$ parallel to the material surface can be significant. It also means that the v_y ion velocity distribution at the magnetic presheath edge may be non-Maxwellian, which gives a non-zero contribution of the viscosity term $m_i n \langle w_y w_x \rangle$. In addition to that, it should be mentioned that the radial viscosity in the bulk plasma is predominantly turbulent, so it is very likely that it is quite significant also at the magnetic presheath edge. However, in the standard version of the B2.5 code these contributions to the momentum fluxes at the material surfaces are not taken into account and even P_{mx} is calculated very approximately. Thus neglecting P_{my} and P_{mz} , the momentum flux at the magnetic presheath edge can be written as:

$$P_{mxMPSE} = \left(\rho_{MPSE} u_{\parallel MPSE}^2 + p_{iMPSE} \right) \sin \theta - m_i c_{sMPSE} D_{turb} \partial_x n_{MPSE}, \quad (136)$$

where $\rho_{MPSE} = m_i n_{MPSE}$ is the ion mass density and $p_{iMPSE} = n_{MPSE} k_B T_{iMPSE}$ is the ion pressure, both calculated at the magnetic presheath edge. The last term on the right-hand side of equation (136) is a phenomenological turbulent momentum flux. In the case of a very small grazing angle θ , this term is much larger than the classical term. Comparison of equation (136) with equations (70) and (93) shows that equation (136) cannot be implemented directly in the standard expressions for the momentum sources. It includes more complicated non-linear dependence on the plasma parameters than equations for the momentum sources (70) and (93), so that they cannot accurately describe the boundary values of the momentum fluxes. It is necessary to make some additional approximations in order to implement equation (136) in the standard expressions for the momentum sources. Assuming that $T_{iMPSE} \approx T_{eMPSE}$, $u_{\parallel MPSE} \approx c_{sMPSE}$ and using the same approximation for the turbulent transport as in the upper analysis of the particle flux, the momentum flux at the magnetic presheath edge can be written approximately as:

$$P_{mxMPSE} \approx \frac{3}{2} \rho_{MPSE} u_{\parallel MPSE}^2 \sin \theta + \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE} \rho_{MPSE} c_{sMPSE}^2. \quad (137)$$

There are several possible choices of the free input parameters C_{mj} and all of them give very approximate values of the boundary sources. The first possibility is to choose $C_{m0} = P_{mxMPSE}$, where P_{mxMPSE} is calculated by using the initial or some other (e.g. experimental) specific values of the ion density, electron and ion temperature, and all other $C_{mj} = 0$. This possibility will be probably the best one in the case of a very small grazing angle θ , i.e., for $\theta \lesssim 5^\circ$, if the value of the momentum flux is provided from some turbulent transport model (see, e.g., [7] and references therein). The second possibility is to specify the free input parameters in the expression for S_{m2} (see equations (73) and (96)). In this case, assuming that $u_{\parallel MPSE} \approx c_{sMPSE}$ the possible choice is $C_{m1} = \left[\frac{3}{2} + \frac{\left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE}}{\sin \theta} \right] c_{sMPSE}$ and all other $C_{mj} = 0$. The third possibility is to specify the free input parameters in the expression for S_{m3} (see equations (74) and (97)). The fluid velocities $u_{\alpha x}$ in equations (70) and (93) are calculated with the bulk transport equations, which means that, in order to preserve the validity of the fluid model and to a certain extent assure the stability of the ion fluid flow, they should not exceed the thermal velocities, i.e., $u_{\alpha x} \leq v_{th,\alpha} \sin \theta$. Then, assuming

that $u_{\parallel MPSE}^2 \approx c_{sMPSE}^2 \approx 2v_{th,i}^2$, the appropriate choice of the free input parameters can be $C_{m3} = \left[\frac{3}{\sqrt{2}} + \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE} \frac{\sqrt{2}}{\sin \theta} \right] c_{sMPSE}$ and all other $C_{mj} = 0$. In all these cases the initial or some other specific values of the plasma parameters should be used in the calculations of the free input parameters C_{mj} .

3.4 Standard Ion and Electron Heat Sources

The ion heat flux at the magnetic presheath edge, $Q_{ix} = \frac{1}{2}m_i n \langle v^2 v_x \rangle$, can be written as (index *MPSE* is dropped) [2]:

$$Q_{ix} = \frac{1}{2}m_i n u^2 u_x + \frac{5}{2}p_i u_x + \Pi_{xx} u_x + \Pi_{xy} u_y + \Pi_{xz} u_z + q_{ix}, \quad (138)$$

where

$$p_i = \frac{1}{3}m_i n \langle w^2 \rangle = nk_B T_i \quad (139)$$

is the scalar pressure,

$$\Pi_{jk} = m_i n \left\langle w_j w_k - \frac{1}{3} w^2 \delta_{jk} \right\rangle \quad (140)$$

is the anisotropic shear (viscosity) tensor and

$$q_{ix} = \frac{1}{2}m_i n \langle w^2 w_x \rangle \quad (141)$$

is the heat conduction flux. All these quantities should be calculated at the magnetic presheath edge. It is obvious again, as in the analysis of the momentum flux, that the contribution of the non-zero radial drift velocity u_y , i.e., the term $\Pi_{xy} u_y$, in equation (138) may be important. Neglecting terms with u_y and u_z in equation (138) and assuming again that the ion velocity distribution at the magnetic presheath edge is drifting-Maxwellian, the ion heat flux at the magnetic presheath edge can be written approximately as:

$$Q_{ixMPSE} = \left(\frac{1}{2}m_i n_{MPSE} u_{\parallel MPSE}^3 + \frac{5}{2}n_{MPSE} k_B T_{iMPSE} u_{\parallel MPSE} \right)$$

$$+2k_B T_{iMPSE} \Gamma_{th,iMPSE}) \sin \theta - \kappa_{turb}^i \partial_x \hat{T}_{iMPSE}, \quad (142)$$

where $\Gamma_{th,iMPSE} = n_{MPSE} \sqrt{\frac{k_B T_{iMPSE}}{2\pi m_i}}$ is the ion thermal flux at the magnetic presheath edge. The last term on the right-hand side of equation (142) is a phenomenological turbulent ion heat conduction flux, which is much larger than the classical ion heat flux, when the grazing angle θ is very small. The ion heat flux at the material wall, Q_{ixW} , is larger because of the electric potential drop between the magnetic presheath edge and the wall, so that:

$$Q_{ixW} = Q_{ixMPSE} + e (\Phi_{MPSE} - \Phi_W) \Gamma_{xMPSE}. \quad (143)$$

On the other hand, this potential drop reduces the electron heat flux at the wall, Q_{exW} , so that:

$$Q_{exW} = Q_{exMPSE} - e (\Phi_{MPSE} - \Phi_W) \Gamma_{xMPSE}. \quad (144)$$

The electron heat flux at the wall, Q_{exW} , is given by:

$$Q_{exW} = \sqrt{\frac{2}{\pi}} n_{eMPSE} v_{th,eMPSE} k_B T_{eMPSE} \exp \left[\frac{e (\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}} \right] \sin \theta - \kappa_{turb}^e \partial_x \hat{T}_{eMPSE}, \quad (145)$$

where $n_{eMPSE} = n_{MPSE}$ in a case of a two-component plasma. The second term on the right-hand side of equation (145) is a phenomenological turbulent electron heat conduction flux, which is much larger than the classical electron heat flux, when the grazing angle θ is very small. Since the boundary sources include the collisionless region with the electric potential drop, its effect on the charged particle heat fluxes must be taken into account. Summing of equations (143) and (144) gives:

$$Q_{ixW} + Q_{exW} = Q_{ixMPSE} + Q_{exMPSE}, \quad (146)$$

which is just an energy conservation law. In the expressions for the boundary heat sources it is possible to use charged particle heat fluxes calculated at the material wall or at the magnetic presheath edge, but in any case the energy conservation law (146) must be satisfied. In this example, the expressions for the boundary heat sources will be calculated with the charged particle heat fluxes calculated at the material wall.

Expressions for the ion and electron heat flux (142)-(145) include more complicated non-linear dependence on the plasma parameters than the standard expressions for the boundary heat sources presented in the previous section (see equations (75), (80), (98) and (103)). Thus additional approximations in equations (142)-(145) are necessary in order to apply them in the standard expressions for the boundary heat sources.

In the absence of the turbulent transport model it is very difficult to make good approximations for the turbulent ion and electron heat flux. In both cases it can be assumed that (indices i , e and $MPSE$ are dropped):

$$Q_{x(turb)} \approx \frac{\kappa_{turb}}{\lambda_{Tx}} k_B T \approx 2k_B T \Gamma_{xturb} \approx 2k_B T \frac{D_{turb}}{\lambda_{nx}} n, \quad (147)$$

where λ_{Tx} and λ_{nx} are characteristic decay lengths for the temperature and density respectively. Assuming $\lambda_{Tx} \approx \frac{\lambda_{nx}}{2} \approx \frac{\lambda_{MPSE}}{2}$, which gives $\kappa_{turb} \approx n D_{turb}$, and with the upper estimate for $\frac{D_{turb}}{\lambda_{MPSE}}$ (see equation (126)), a relatively simple expression for the turbulent heat flux is obtained:

$$Q_{i,e,MPSE(turb)} \approx 2 \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE} c_{sMPSE} n_{MPSE} k_B T_{i,eMPSE}. \quad (148)$$

Approximate expression for the turbulent heat flux (148) can be applied in the standard expression for the boundary heat source. Assuming $T_{iMPSE} \approx T_{eMPSE}$ and $u_{\parallel MPSE} \approx c_{sMPSE} \approx \sqrt{2} v_{th,iMPSE}$ in the classical ion heat flux, the approximate expression for the ion heat flux at the wall is:

$$Q_{ixW} \approx \left[\frac{7}{2} + \frac{1}{\sqrt{\pi}} + \frac{e(\Phi_{MPSE} - \Phi_W)}{k_B T_{eMPSE}} \right] n_{MPSE} u_{\parallel MPSE} k_B T_{iMPSE} \sin \theta$$

$$+ \left[2 + \frac{e(\Phi_{MPSE} - \Phi_W)}{k_B T_{eMPSE}} \right] \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE} n_{MPSE} c_{sMPSE} k_B T_{iMPSE}. \quad (149)$$

The approximate expression for the electron heat flux at the wall is:

$$Q_{exW} \approx \sqrt{\frac{2}{\pi}} n_{eMPSE} v_{th,eMPSE} k_B T_{eMPSE} \exp \left[\frac{e(\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}} \right] \sin \theta \\ + 2 \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE} n_{MPSE} c_{sMPSE} k_B T_{eMPSE}. \quad (150)$$

In order to specify the numerical values of the free input parameters C_{hij} and C_{hej} , it is also necessary to obtain the numerical value for the normalized potential fall $\frac{e(\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}}$ in equations (149) and (150). It can be obtained directly from the experiment, e.g., from measurements of the spatial profiles of the electric potential with emissive probes and electron temperature with Langmuir (collecting) probes, or calculated from the measured electric current to the divertor plate, $I_d = \pm j_d A_d$. The total electric current density at the divertor plate, j_d , can be written as:

$$j_d = j_{iMPSEx} + j_{eMPSEx} - en_{MPSE} u_{\parallel MPSE} \sin \theta - \frac{(1 - \gamma_{se})}{\sqrt{2\pi}} e n_{eMPSE} v_{th,eMPSE} \exp \left[\frac{e(\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}} \right] \sin \theta, \quad (151)$$

where γ_{se} is the secondary electron emission coefficient and it is assumed that the turbulent transport is ambipolar, i.e., $D_{turb}^i = D_{turb}^e$, so that its net contribution to the total electric current is zero. The secondary electron emission is not taken into account in the expression for the electron current (18), although it can be important in some practical cases. In such cases it is necessary to include the correction due to the secondary electron emission in the expression for the electric charge source, which is presented in the text below. If the secondary electron emission varies along the material surface, so that the effective secondary electron emission coefficient is not constant, this effect must be explicitly taken into account in the expression for the electron current (18). Equation (151) is applicable only if the grazing angle θ is not very small, i.e., if $\theta \gtrsim 5^\circ$. In the case of a very small grazing angle θ it may be assumed that $\frac{e(\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}} \approx 0$ in equations (149) and (150)

or the experimental value for that specific case should be used. Considering $n_{eMPSE} = n_{MPSE}$, equation (151) can be rewritten as:

$$\frac{e(\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}} = \ln \left[\frac{\sqrt{2\pi} u_{\parallel MPSE}}{(1 - \gamma_{se}) v_{th,eMPSE}} - \frac{j_d}{\frac{(1-\gamma_{se})}{\sqrt{2\pi}} e n_{MPSE} v_{th,eMPSE} \sin \theta} \right], \quad (152)$$

where some specific values of the plasma parameters, e.g., the initial values in the simulation ($n_{i,e}^{(0)}$, $\hat{T}_{i,e}^{(0)}$, $u_D^{(0)}$) or experimental values, should be used in the right-hand side of equation (152). For example, it can be assumed that $u_{\parallel MPSE} \approx c_{sMPSE}$ and for the plasma density at the magnetic presheath edge $n_{MPSE} \approx n_{i,e}^{(0)} \exp \left(\frac{e(\Phi_{MPSE} - \Phi_{\infty}^{(0)})}{k_B T_e^{(0)}} \right) \approx n_{i,e}^{(0)} \exp \left(-\frac{1}{2} \right)$, because the potential fall in the presheath, which is approximately $-\frac{k_B T_e^{(0)}}{2e}$, accelerates the ions from their thermal velocities up to the ion sound velocity. In a case of electrically floating divertor plates, $j_d = 0$, so that:

$$\frac{e(\Phi_W^f - \Phi_{MPSE}^f)}{k_B T_{eMPSE}} = \ln \left[\frac{\sqrt{2\pi} u_{\parallel MPSE}}{(1 - \gamma_{se}) v_{th,eMPSE}} \right]. \quad (153)$$

Assuming $u_{\parallel MPSE} \approx c_{sMPSE}$, equation (153) can be rewritten as:

$$\frac{e(\Phi_W^f - \Phi_{MPSE}^f)}{k_B T_{eMPSE}} = \frac{1}{2} \ln \left[\frac{2\pi m_e \left(1 + \frac{T_{iMPSE}}{T_{eMPSE}} \right)}{m_i (1 - \gamma_{se})^2} \right]. \quad (154)$$

Thus if $T_{iMPSE} \approx T_{eMPSE}$ and $\gamma_{se} \ll 1$,

$$\frac{e(\Phi_W^f - \Phi_{MPSE}^f)}{k_B T_{eMPSE}} \approx \frac{1}{2} \ln \left[\frac{4\pi m_e}{m_i} \right]. \quad (155)$$

The appropriate choice of the free input parameters is $C_{hi2} = [2 + \frac{e(\Phi_{MPSE} - \Phi_W)}{k_B T_{eMPSE}}] \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE}$, $C_{hi3} = \frac{7}{2} + \frac{1}{\sqrt{\pi}} + \frac{e(\Phi_{MPSE} - \Phi_W)}{k_B T_{eMPSE}}$ and all other

$C_{hij} = 0$ for the ion heat source and $C_{he3} = \sqrt{\frac{2}{\pi}} \exp \left[\frac{e(\Phi_W - \Phi_{MPSE})}{k_B T_{eMPSE}} \right]$,
 $C_{he4} = 2 \left(\frac{\delta n}{\langle n \rangle} \right)_{MPSE}$ and all other $C_{hej} = 0$ for the electron heat source.

3.5 Standard Electric Charge Sources

The electric charge source in each boundary cell is given by equation:

$$S_{ch} = -j_d S_0 = J_{ex} - J_{ix}, \quad (156)$$

where j_d is given by equation (151) and S_0 is given by equation (6). In the general case, when $j_d \neq 0$, the appropriate choice of the free input parameters is $C_{ch6} = 1$, $C_{ch7} = 1 - \gamma_{se}$ and all other $C_{chj} = 0$. The non-zero numerical values of the electric potential at the divertor plates, $\Phi_d^{(0)}$, due to the electric bias, can be specified in the input file `b2ai.dat`, which describes the initial plasma state ($\Phi_d^{(0)} = 0$ is a default value). In a case of electrically floating divertor plates, $j_d = 0$, so that $S_{ch} = 0$ in each boundary cell. It can be obtained by choosing all $C_{chj} = 0$.

3.6 Conclusion

It is obvious from this relatively simple example that only the expressions for the particle sources, i.e., the numerical implementation of the boundary conditions for the continuity equation, can adequately reproduce the physical model of the magnetized boundary plasma, after some minor corrections. It is necessary to include in the source code some more complicated non-linear dependences of the boundary sources on the macroscopic plasma parameters, so that the boundary conditions for the higher moments' fluid equations can be correctly applied. The approximate fluid model used in the upper example, which is based on the assumption that the ion velocity distribution at the magnetic presheath edge is drifting-Maxwellian distribution, can be a good starting point in the effort to improve the standard expressions for the boundary sources.

4 Boundary Sources Written by D. P. Coster

4.1 Introduction

In order to improve the numerical implementation of the boundary conditions, an alternative method for calculating the boundary sources has been developed by D. P. Coster from IPP Garching. Their dependence on the plasma parameters calculated with the bulk plasma equations is mostly the same as for the standard boundary sources, but the way how the free input parameters are specified and used in the calculations of the coefficients $S_{pj\alpha}$, $S_{mj\alpha}$, S_{hij} , S_{hej} and S_{ch} is different. The B2.5 user can choose between the standard (default) version and this version of the boundary sources by specifying the numerical value of the input parameter `boundary_namelist` in the `b2mn.dat` input file (default is zero). In order to use the non-standard version of the boundary sources, a line of text: `'b2stbc.boundary_namelist'1'`, should be added into the `b2mn.dat` file. The exact specifications of the types of the boundary conditions and numerical values of the free input parameters, which are used in the calculations of the non-standard boundary sources, should be written in the `b2.boundary.parameters` file. Following specifications of the boundary sources and corresponding labels should be written in the `b2.boundary.parameters` file:

- `&BOUNDARY` (the first line or beginning of the file)
- `NBC=6`, (defines the number of the boundary conditions or the number of the boundary regions)
- `BCCHAR= 'S ', 'S', 'S', 'N', 'W', 'E'`, (`BCCHAR(1...NBC)` defines the boundary as `'S', 'N', 'W', 'E'`)
- `BCPOS=...`, (`BCPOS(1...NBC)` defines the position of the boundary either in ix (`'W'` or `'E'`) or iy (`'S'` or `'N'`))
- `BCSTART=...`, (`BCSTART(1...NBC)` defines the start of the boundary region either in ix (`'S'` or `'N'`) or iy (`'W'` or `'E'`))
- `BCEND=...`, (`BCEND(1...NBC)` defines the end of the boundary region either in ix (`'S'` or `'N'`) or iy (`'W'` or `'E'`))

- BCENE=..., (BCENE(1...NBC)=1-10 defines the type of the boundary condition for the electron energy equation)
- BCENI=..., (BCENI(1...NBC)=1-10 defines the type of the boundary condition for the ion energy equation)
- BCPOT=..., (BCPOT(1...NBC)=1-9 defines the type of the boundary condition for the electric potential equation)
- ENEPAR=..., (ENEPAR(1...NBC, 1) and/or ENEPAR(1...NBC, 2) are free input parameters for the boundary condition for the electron energy equation, chosen according to the corresponding BCENE(1...NBC))
- ENIPAR=..., (ENIPAR(1...NBC, 1) and/or ENIPAR(1...NBC, 2) are free input parameters for the boundary condition for the ion energy equation, chosen according to the corresponding BCENI(1...NBC))
- POTPAR=..., (POTPAR(1...NBC, 1) and/or POTPAR(1...NBC, 2) are free input parameters for the boundary condition for the electric potential equation, chosen according to the corresponding BCPOT(1...NBC))
- BCCON=..., (BCCON(1...NS, 1...NBC)=1-10 defines the type of the boundary condition for the continuity equation for species IS=1...NS)
- CONPAR=..., (CONPAR(1...NS, 1...NBC, 1) and/or CONPAR(1...NS, 1...NBC, 2) are free input parameters for the boundary condition for the continuity equation for species IS=1...NS, chosen according to the corresponding BCCON(1...NS, 1...NBC))
- BCMOM=..., (BCMOM(1...NS, 1...NBC)=1-11 defines the type of the boundary condition for the momentum equation for species IS=1...NS)
- MOMPARG=..., (MOMPARG(1...NS, 1...NBC, 1) and/or MOMPARG(1...NS, 1...NBC, 2) are free input parameters for the boundary condition for the momentum equation for species IS=1...NS, chosen according to the corresponding BCMOM(1...NS, 1...NBC))
- GAMMAI=..., GAMMAE=..., (ion adiabatic coefficient, secondary electron emission coefficient)
- other labels for various input options, which are not implemented yet (e.g., LBNDUSR=t (true) calls the user-routine `bndusr.f`, otherwise LBNDUSR=f (false)),

- / (the last line or end of the file).

If the value of `boundary_namelist` is zero or the upper line of text is not added, the standard boundary sources will be calculated.

This report describes the non-standard boundary sources included in the version of `b2stbc.f` file from September 20, 1999 (see also [18]). Some possible input parameter options for the non-standard boundary sources have not been implemented yet. There are also several certain or possible programming errors in the source code, in addition to some theoretically questionable and/or insufficiently justified calculations. These possible errors are also included in the analytical expressions presented in this report. Since this part of the B2.5 code is still not completely developed and debugged, these insufficiencies are not analysed in detail in this report. However, it can be concluded that these new boundary sources do not include significant qualitative improvements of the standard boundary sources, or in some cases they can even less accurately describe the boundary conditions compared with the standard boundary sources.

4.2 Analytical Expressions for the "New" Boundary Sources

4.2.1 Physical Parameters and Numerical Constants

Several new numerical constants, which are related to the physical velocities and particle density are defined. Their numerical values are the following:

$$EPSLON = 10^{-10} \equiv \epsilon, \quad (157)$$

$$NE00 = 10^{20} \equiv \mathcal{N}_e, \quad (158)$$

$$UP00 = \sqrt{\frac{100\text{eV}}{m_p}} \quad 97873 \frac{m}{s} \equiv \mathcal{U}_p, \quad (159)$$

$$UBIG = \frac{UP00}{EPSLON} \equiv \mathcal{U} \quad (160)$$

and

$$VBIG = 10^{-2}UBIG \equiv \mathcal{V}. \quad (161)$$

Their physical background is not explained in the comments. Anomalous diffusion coefficients D_{an}^α and D_{ap}^α , viscosities η_a^α , heat conductivities κ_a^e and κ_a^i and electrical conductivity σ_a are also used in the calculations of the boundary sources. s_x is the area of the face between the cells (ix, iy) and (ix+1,iy) and s_y between the cells (ix,iy) and (ix,iy+1). In the expressions for the boundary sources they are calculated at the boundary cells. u_α is the poloidal component of the drift velocity of the atomic species α , $u_{D\alpha}$ is the poloidal component of the $\mathbf{E} \times \mathbf{B}$ drift velocity and $b_x = \frac{B_x}{B}$ is the magnetic field pitch. The ion sound velocity for the multispecies plasma is defined as:

$$c_s = \sqrt{\frac{p}{\rho}}, \quad (162)$$

where p is the total pressure and $\rho = m_p \sum_\alpha A_\alpha n_\alpha$ is the total mass density. The ion sound velocity of the species α is defined as:

$$c_{s\alpha} = \sqrt{\frac{\gamma_i \hat{T}_i + Z_\alpha^2 r_\alpha \hat{T}_e}{m_p A_\alpha}}, \quad (163)$$

where $\gamma_i = \frac{c_{pi}}{c_{vi}}$ is the ratio of the ion specific heats or the ion adiabatic coefficient ($\gamma_i = \frac{5}{3}$ for the adiabatic fluid model, $\gamma_i = 1$ for the isothermal fluid model, $\gamma_i = 3$ for the one-dimensional kinetic model [4]) and $r_\alpha = \frac{n_\alpha}{n_e}$. C_{leak} is a dimensionless leakage coefficient. γ_{se} is the secondary electron emission coefficient. The other physical parameters are described in the following text, when they are used in the expressions for the boundary sources.

4.2.2 Boundary Conditions for Continuity Equations

BCCON=1 ... Prescribe the Value of the Density

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv N_\alpha$$

- SOUTH and NORTH boundary:

$$S_{p0\alpha} = \mathcal{V} N_\alpha s_y, \quad (164)$$

$$S_{p1\alpha} = -\mathcal{V} s_y, \quad (165)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{V} s_y (N_\alpha - n_\alpha). \quad (166)$$

- WEST and EAST boundary:

$$S_{p0\alpha} = \mathcal{U} N_\alpha s_x, \quad (167)$$

$$S_{p1\alpha} = -\mathcal{U} s_x, \quad (168)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{U} s_x (N_\alpha - n_\alpha). \quad (169)$$

BCCON=2 ... Prescribe the Value of the Gradient of the Density

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv \partial_x N_\alpha, \partial_y N_\alpha$$

- SOUTH boundary:

$$l_y = \frac{1}{2} (h_{y1(S)} + h_{y1(S+1)}) , \quad (170)$$

$$S_{p0\alpha} = \mathcal{V} s_y [(-\partial_y N_\alpha) l_y + n_\alpha] , \quad (171)$$

$$S_{p1\alpha} = -\mathcal{V} s_y , \quad (172)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{V} s_y (-\partial_y N_\alpha) l_y . \quad (173)$$

- NORTH boundary:

$$l_y = \frac{1}{2} (h_{y1(N-1)} + h_{y1(N)}) , \quad (174)$$

$$S_{p0\alpha} = \mathcal{V} s_y [(\partial_y N_\alpha) l_y + n_\alpha] , \quad (175)$$

$$S_{p1\alpha} = -\mathcal{V} s_y , \quad (176)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{V} s_y (\partial_y N_\alpha) l_y . \quad (177)$$

- WEST boundary:

$$l_x = \frac{1}{2} (h_{x(W)} + h_{x(W+1)}) , \quad (178)$$

$$S_{p0\alpha} = \mathcal{U} s_x [(-\partial_x N_\alpha) l_x + n_\alpha] , \quad (179)$$

$$S_{p1\alpha} = -\mathcal{U} s_x , \quad (180)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{U} s_x (-\partial_x N_\alpha) l_x . \quad (181)$$

- EAST boundary:

$$l_x = \frac{1}{2} (h_{x(E-1)} + h_{x(E)}) , \quad (182)$$

$$S_{p0\alpha} = \mathcal{U} s_x [(\partial_x N_\alpha) l_x + n_\alpha] , \quad (183)$$

$$S_{p1\alpha} = -\mathcal{U} s_x , \quad (184)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{U} s_x (\partial_x N_\alpha) l_x . \quad (185)$$

BCCON=3 ... Sheath Conditions (Only for the WEST and EAST Boundary)

$$S_{p0\alpha} = \mathcal{U} n_\alpha s_x , \quad (186)$$

$$S_{p1\alpha} = -\mathcal{U} s_x , \quad (187)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = 0.^{(*)} \quad (188)$$

^(*)WARNING: PROBABLY ERROR - $S_{p1\alpha}$ should have some other value, so that $S_{p\alpha} \neq 0$!

BCCON=4 ... Prescribe the Value of the Density "Weakly", Using as the Control Parameter Some Prescribed Velocity

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv N_\alpha,$$

$$\text{CONPAR}(\text{IS}, \text{IB}, 2) \equiv U_\alpha$$

- for all boundaries (N_α and U_α are different at each boundary):

$$S_{p0\alpha} = N_\alpha U_\alpha s, \quad (189)$$

$$S_{p1\alpha} = -U_\alpha s, \quad (190)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = U_\alpha s (N_\alpha - n_\alpha). \quad (191)$$

BCCON=5 ... Prescribe the Value of the Particle Flux per Unit Area

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv \Gamma_{\alpha x}, \Gamma_{\alpha y}$$

- SOUTH and NORTH boundary:

$$S_{p0\alpha} = \Gamma_{\alpha y} s_y, \quad (192)$$

$$S_{p1\alpha} = 0, \quad (193)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \Gamma_{\alpha y} s_y. \quad (194)$$

- WEST and EAST boundary:

$$S_{p0\alpha} = \Gamma_{\alpha x} s_x, \quad (195)$$

$$S_{p1\alpha} = 0, \quad (196)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \Gamma_{\alpha x} s_x. \quad (197)$$

BCCON=6 ... Prescribe the Value of the Total Particle Flux for a Constant Density (Only for the NORTH and SOUTH Boundary)

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv \mathcal{S}_n$$

- NORTH boundary:

$$s_1 = \sum_j^N \frac{s_{yj}}{\frac{1}{2} (h_{y1(j,N-1)} + h_{y1(j,N)})}, \quad (198)$$

$$s_2 = \sum_j^N \frac{n_{\alpha j} s_{yj}}{\frac{1}{2} (h_{y1(j,N-1)} + h_{y1(j,N)})}, \quad (199)$$

- SOUTH boundary:

$$s_1 = \sum_j^S \frac{s_{yj}}{\frac{1}{2} (h_{y1(j,S)} + h_{y1(j,S+1)})}, \quad (200)$$

$$s_2 = \sum_j^S \frac{n_{\alpha j} s_{yj}}{\frac{1}{2} (h_{y1(j,S)} + h_{y1(j,S+1)})}, \quad (201)$$

where $\sum_j^N$ and $\sum_j^S$ are summations over the boundary cells at the NORTH and SOUTH boundary respectively. The particle sources are defined as:

$$S_{p0\alpha} = \mathcal{V} s_y (\mathcal{S}_n / D_{an} + s_2) / s_1, \quad (202)$$

$$S_{p1\alpha} = -\mathcal{V} s_y, \quad (203)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{V} s_y [\mathcal{S}_n / (D_{an} s_1) + s_2 / s_1 - n_\alpha]. \quad (204)$$

BCCON=7 ... Prescribe the Value of the Density as a Function of Other Plasma Parameters (Not Yet Available)

BCCON=8 ... Prescribe the Value of the Total Particle Flux with a Constant Flux Density

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv \mathcal{S}_{\Gamma_x}, \mathcal{S}_{\Gamma_y}$$

- SOUTH and NORTH boundary:

$$s_{tot} = \sum_j^{S \text{ or } N} s_{yj}, \quad (205)$$

$$S_{p0\alpha} = (\mathcal{S}_{\Gamma_y} / s_{tot}) s_y, \quad (206)$$

$$S_{p1\alpha} = 0, \quad (207)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_\alpha = (\mathcal{S}_{\Gamma_y}/s_{tot}) s_y. \quad (208)$$

- WEST and EAST boundary:

$$s_{tot} = \sum_j^{W \text{ or } E} s_{xj}, \quad (209)$$

$$S_{p0\alpha} = (\mathcal{S}_{\Gamma_x}/s_{tot}) s_x, \quad (210)$$

$$S_{p1\alpha} = 0, \quad (211)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_\alpha = (\mathcal{S}_{\Gamma_x}/s_{tot}) s_x. \quad (212)$$

**BCCON=9 ... Prescribe the Value of the Decay Length for the Density
(Only for the SOUTH and NORTH Boundary)**

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv l_{n_\alpha} = n_\alpha / \partial_y n_\alpha$$

$$S_{p0\alpha} = 0, \quad (213)$$

$$S_{p1\alpha} = - \left[D_{an_\alpha} + D_{ap_\alpha} \left(Z_\alpha \hat{T}_e + \hat{T}_i \right) \right] s_y / l_{n_\alpha}, \quad (214)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha}n_\alpha = - \left[D_{an_\alpha} + D_{ap_\alpha} \left(Z_\alpha \hat{T}_e + \hat{T}_i \right) \right] s_y n_\alpha / l_{n_\alpha}. \quad (215)$$

BCCON=10 ... Prescribe the Leakage Coefficient (Leakage Option for the Density, Only for the SOUTH and NORTH Boundary)

$$\text{CONPAR}(\text{IS}, \text{IB}, 1) \equiv C_{leak}$$

$$S_{p0\alpha} = 0, \quad (216)$$

$$S_{p1\alpha} = \frac{V}{h_{y1}} \sqrt{\frac{Z_\alpha \hat{T}_e + \hat{T}_i}{m_p A_\alpha}} C_{leak}, \quad (217)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \frac{V}{h_{y1}} \sqrt{\frac{Z_\alpha \hat{T}_e + \hat{T}_i}{m_p A_\alpha}} C_{leak} n_\alpha. \quad (218)$$

- total pumping rate of the species α at the SOUTH and NORTH boundary (summation over all boundary cells, added by H. B rbaumer):

$$R_{p\alpha} = \sum_j^{S \text{ or } N} S_{p\alpha j} = \sum_j^{S \text{ or } N} S_{p1\alpha j} n_{\alpha j}. \quad (219)$$

4.2.3 Boundary Conditions for Momentum Equations

BCMOM=1 ... Prescribe the Value of the Parallel Velocity

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv U_\alpha$$

- SOUTH and NORTH boundary:

$$S_{m0\alpha} = \mathcal{V}\mathcal{N}_e A_\alpha U_\alpha s_y, \quad (220)$$

$$S_{m1\alpha} = -\mathcal{V}\mathcal{N}_e A_\alpha s_y, \quad (221)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V}\mathcal{N}_e A_\alpha s_y (U_\alpha - u_\alpha). \quad (222)$$

- WEST and EAST boundary:

$$S_{m0\alpha} = \mathcal{U}\mathcal{N}_e A_\alpha U_\alpha s_x, \quad (223)$$

$$S_{m1\alpha} = -\mathcal{U}\mathcal{N}_e A_\alpha s_x, \quad (224)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{U}\mathcal{N}_e A_\alpha s_x (U_\alpha - u_\alpha). \quad (225)$$

BCMOM=2 ... Prescribe the Gradient of the Parallel Velocity

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv \partial_x U_\alpha, \quad \partial_y U_\alpha$$

- NORTH boundary:

$$\bar{n}_\alpha = \frac{1}{2} (n_{\alpha(ix,N)} + n_{\alpha(ix+1,N)}), \quad (226)$$

$$l_y = \frac{1}{4} (h_{y1(ix,N-1)} + h_{y1(ix+1,N-1)}), \quad (227)$$

$$\widehat{s} = \frac{1}{2} (b_{x(ix, N-1)} + b_{x(ix+1, N-1)}) , \quad (228)$$

$$\widehat{c} = \frac{1}{2} \left(\sqrt{1 - b_{x(ix, N-1)}^2} + \sqrt{1 - b_{x(ix+1, N-1)}^2} \right) , \quad (229)$$

$$S_{m0\alpha} = \mathcal{V} s_y A_\alpha \bar{n}_\alpha [(\partial_y U_\alpha) l_y + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s}] , \quad (230)$$

$$S_{m1\alpha} = -\mathcal{V} s_y A_\alpha \bar{n}_\alpha , \quad (231)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V} s_y A_\alpha \bar{n}_\alpha [(\partial_y U_\alpha) l_y + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s} - u_\alpha] .^{(*)} \quad (232)$$

^(*)WARNING: PROBABLY ERROR - THE AUTHOR PROBABLY WANTED TO GET THE RESULT:

$$S_{m\alpha} = \mathcal{V} s_y A_\alpha \bar{n}_\alpha (\partial_y U_\alpha) l_y .$$

- SOUTH boundary:

$$\bar{n}_\alpha = \frac{1}{2} (n_{\alpha(ix, S)} + n_{\alpha(ix+1, S)}) , \quad (233)$$

$$l_y = \frac{1}{4} (h_{y1(ix, S+1)} + h_{y1(ix+1, S+1)}) , \quad (234)$$

$$\widehat{s} = \frac{1}{2} (b_{x(ix, S+1)} + b_{x(ix+1, S+1)}) , \quad (235)$$

$$\widehat{c} = \frac{1}{2} \left(\sqrt{1 - b_{x(ix, S+1)}^2} + \sqrt{1 - b_{x(ix+1, S+1)}^2} \right), \quad (236)$$

$$S_{m0\alpha} = \mathcal{V}_{sy} A_\alpha \overline{n}_\alpha \left[-(\partial_y U_\alpha) l_y + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s} \right], \quad (237)$$

$$S_{m1\alpha} = -\mathcal{V}_{sy} A_\alpha \overline{n}_\alpha, \quad (238)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V}_{sy} A_\alpha \overline{n}_\alpha \left[-(\partial_y U_\alpha) l_y + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s} - u_\alpha \right].^{(*)} \quad (239)$$

^(*)WARNING: PROBABLY ERROR - THE AUTHOR PROBABLY WANTED TO GET THE RESULT:

$$S_{m\alpha} = -\mathcal{V}_{sy} A_\alpha \overline{n}_\alpha (\partial_y U_\alpha) l_y.$$

- WEST boundary:

$$\overline{n}_\alpha = \frac{1}{2} \left(n_{\alpha(W, iy)} + n_{\alpha(W+1, iy)} \right), \quad (240)$$

$$l_x = \frac{1}{4} \left(h_{x(W, iy)} + h_{x(W+1, iy)} \right), \quad (241)$$

$$\widehat{s} = \frac{1}{2} \left(b_{x(W, iy)} + b_{x(W+1, iy)} \right), \quad (242)$$

$$\widehat{c} = \frac{1}{2} \left(\sqrt{1 - b_{x(W, iy)}^2} + \sqrt{1 - b_{x(W+1, iy)}^2} \right), \quad (243)$$

$$S_{m0\alpha} = \mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha [- (\partial_x U_\alpha) l_x + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s}], \quad (244)$$

$$S_{m1\alpha} = -\mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha, \quad (245)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha [- (\partial_x U_\alpha) l_x + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s} - u_\alpha]^{(*)} \quad (246)$$

(*)WARNING: PROBABLY ERROR - THE AUTHOR PROBABLY WANTED TO GET THE RESULT:

$$S_{m\alpha} = -\mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha (\partial_x U_\alpha) l_x.$$

- EAST boundary:

$$\bar{n}_\alpha = \frac{1}{2} (n_{\alpha(E-1, iy)} + n_{\alpha(E, iy)}), \quad (247)$$

$$l_x = \frac{1}{4} (h_{x(E-1, iy)} + h_{x(E, iy)}), \quad (248)$$

$$\widehat{s} = \frac{1}{2} (b_{x(E-1, iy)} + b_{x(E, iy)}), \quad (249)$$

$$\widehat{c} = \frac{1}{2} \left(\sqrt{1 - b_{x(E-1, iy)}^2} + \sqrt{1 - b_{x(E, iy)}^2} \right), \quad (250)$$

$$S_{m0\alpha} = \mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha [(\partial_x U_\alpha) l_x + (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s}], \quad (251)$$

$$S_{m1\alpha} = -\mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha, \quad (252)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha [(\partial_x U_\alpha) l_x + (u_\alpha - \hat{c} u_{D\alpha}) / \hat{s} - u_\alpha]^{(*)} \quad (253)$$

^(*)WARNING: PROBABLY ERROR - THE AUTHOR PROBABLY WANTED TO GET THE RESULT:

$$S_{m\alpha} = \mathcal{V}_{s_x} A_\alpha \bar{n}_\alpha (\partial_x U_\alpha) l_x.$$

BCMOM=3 ... Sheath Conditions (Only for the WEST and EAST Boundary), Mach Number as Input

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv \mathcal{M}_{\alpha 1},$$

$$\text{MOMPAR}(\text{IS}, \text{IB}, 2) \equiv \mathcal{M}_{\alpha 2}$$

- WEST boundary:

IF ($\mathcal{M}_{\alpha 2} < 0.5$) THEN:

$$u_{out} = b_x \mathcal{M}_{11} c_s, \quad (254)$$

$$u_{p\alpha} = \frac{u_{out}}{b_x} + u_{D1}, \quad (255)$$

ELSE ($\mathcal{M}_{\alpha 2} \geq 0.5$):

$$u_{test\alpha} = u_{\alpha(W+2, iy)} \left(1 + \frac{h_{x(W+1, iy)}}{h_{x(W+2, iy)}} \right) - u_{\alpha(W+3, iy)} \frac{h_{x(W+1, iy)}}{h_{x(W+2, iy)}}, \quad (256)$$

$$u_{out\alpha} = \max(\mathcal{M}_{\alpha 1} c_{s\alpha}, -u_{test\alpha}), \quad (257)$$

$$u_{p\alpha} = u_{out\alpha} + u_{D1}, \quad (258)$$

where u_{D1} is the poloidal component of the $\mathbf{E} \times \mathbf{B}$ drift velocity.

$$S_{m0\alpha} = \mathcal{U} A_{\alpha} m_p n_{\alpha} s_x (-u_{p\alpha}), \quad (259)$$

$$S_{m1\alpha} = -\mathcal{U} A_{\alpha} m_p n_{\alpha} s_x, \quad (260)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_{\alpha} = \mathcal{U} A_{\alpha} m_p n_{\alpha} s_x (-u_{p\alpha} - u_{\alpha}). \quad (261)$$

- EAST boundary:

IF ($\mathcal{M}_{\alpha 2} < 0.5$) THEN:

$$u_{out} = b_x \mathcal{M}_{11} c_s, \quad (262)$$

$$u_{p\alpha} = \frac{u_{out}}{b_x} - u_{D1}, \quad (263)$$

ELSE ($\mathcal{M}_{\alpha 2} \geq 0.5$):

$$u_{test\alpha} = u_{\alpha(E-1, iy)} \left(1 + \frac{h_{x(E-1, iy)}}{h_{x(E-2, iy)}} \right) - u_{\alpha(E-2, iy)} \frac{h_{x(E-1, iy)}}{h_{x(E-2, iy)}}, \quad (264)$$

$$u_{out\alpha} = \max(\mathcal{M}_{\alpha 1} c_{s\alpha}, u_{test\alpha}), \quad (265)$$

$$u_{p\alpha} = u_{out\alpha} - u_{D1}, \quad (266)$$

where u_{D1} is the poloidal component of the $\mathbf{E} \times \mathbf{B}$ drift velocity.

$$S_{m0\alpha} = \mathcal{U} A_{\alpha} m_p n_{\alpha} s_x u_{p\alpha}, \quad (267)$$

$$S_{m1\alpha} = -\mathcal{U} A_{\alpha} m_p n_{\alpha} s_x, \quad (268)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_{\alpha} = \mathcal{U} A_{\alpha} m_p n_{\alpha} s_x (u_{p\alpha} - u_{\alpha}). \quad (269)$$

**BCMOM=4 ... Prescribe the Value of the Velocity "Weakly", Using
as the Control Parameter Some Prescribed Mass Flux**

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv U_{\alpha},$$

$$\text{MOMPAR}(\text{IS}, \text{IB}, 2) \equiv G_{\alpha x}, G_{\alpha y}$$

- SOUTH and NORTH boundary:

$$S_{m0\alpha} = G_{\alpha y} U_{\alpha} s_y, \quad (270)$$

$$S_{m1\alpha} = -G_{\alpha y} s_y, \quad (271)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_{\alpha} = G_{\alpha y} s_y (U_{\alpha} - u_{\alpha}). \quad (272)$$

- WEST and EAST boundary:

$$S_{m0\alpha} = G_{\alpha x} U_{\alpha} s_x, \quad (273)$$

$$S_{m1\alpha} = -G_{\alpha x} s_x, \quad (274)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_{\alpha} = G_{\alpha x} s_x (U_{\alpha} - u_{\alpha}). \quad (275)$$

BCMOM=5 ... Prescribe the Value of the Momentum Flux per Unit Area

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv P_{m\alpha x}, P_{m\alpha y}$$

- SOUTH and NORTH boundary:

$$S_{m0\alpha} = P_{m\alpha y}s_y, \quad (276)$$

$$S_{m1\alpha} = 0, \quad (277)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_\alpha = P_{m\alpha y}s_y. \quad (278)$$

- WEST and EAST boundary:

$$S_{m0\alpha} = P_{m\alpha x}s_x, \quad (279)$$

$$S_{m1\alpha} = 0, \quad (280)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_\alpha = P_{m\alpha x}s_x. \quad (281)$$

BCMOM=6 ... Prescribe the Value of the Total Momentum Flux for a Constant Parallel Velocity (Not Yet Available)

BCMOM=7 ... Prescribe the Density as a Function of the Other Plasma Parameters (Not Yet Available)

BCMOM=8 ... **Limited Shear (Only for the SOUTH and NORTH Boundary)**

- SOUTH boundary:

$$a = \min \left(1, \sqrt{\frac{\widehat{T}_{i(ix,S)} + \widehat{T}_{i(ix+1,S)}}{\widehat{T}_{i(ix,S+1)} + \widehat{T}_{i(ix+1,S+1)}}} \right), \quad (282)$$

$$\overline{n}_\alpha = \frac{1}{2} (n_{\alpha(ix,S+1)} + n_{\alpha(ix+1,S+1)}), \quad (283)$$

$$\widehat{s} = \frac{1}{2} (b_{x(ix,S+1)} + b_{x(ix+1,S+1)}), \quad (284)$$

$$\widehat{c} = \frac{1}{2} (\sqrt{1 - b_{x(ix,S+1)}^2} + \sqrt{1 - b_{x(ix+1,S+1)}^2}), \quad (285)$$

$$S_{m0\alpha} = \mathcal{V} s_y A_\alpha \overline{n}_\alpha a (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s}, \quad (286)$$

$$S_{m1\alpha} = -\mathcal{V} s_y A_\alpha \overline{n}_\alpha, \quad (287)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V} s_y A_\alpha \overline{n}_\alpha [a (u_\alpha - \widehat{c} u_{D\alpha}) / \widehat{s} - u_\alpha]. \quad (288)$$

- NORTH boundary:

$$a = \min \left(1, \sqrt{\frac{\widehat{T}_{i(ix,N-1)} + \widehat{T}_{i(ix+1,N-1)}}{\widehat{T}_{i(ix,N)} + \widehat{T}_{i(ix+1,N)}}} \right), \quad (289)$$

$$\bar{n}_\alpha = \frac{1}{2} (n_{\alpha(ix,N)} + n_{\alpha(ix+1,N)}) , \quad (290)$$

$$\hat{s} = \frac{1}{2} (b_{x(ix,N)} + b_{x(ix+1,N)}) , \quad (291)$$

$$\hat{c} = \frac{1}{2} \left(\sqrt{1 - b_{x(ix,N)}^2} + \sqrt{1 - b_{x(ix+1,N)}^2} \right) , \quad (292)$$

$$S_{m0\alpha} = \mathcal{V} s_y A_\alpha \bar{n}_\alpha a (u_\alpha - \hat{c} u_{D\alpha}) / \hat{s} , \quad (293)$$

$$S_{m1\alpha} = -\mathcal{V} s_y A_\alpha \bar{n}_\alpha , \quad (294)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = \mathcal{V} s_y A_\alpha \bar{n}_\alpha [a (u_\alpha - \hat{c} u_{D\alpha}) / \hat{s} - u_\alpha] . \quad (295)$$

BCMOM=9 ... Prescribe the Value of the Total Momentum Flux with Constant Flux Density

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv \mathcal{S}_{m\alpha x}, \mathcal{S}_{m\alpha y}$$

- SOUTH and NORTH boundary:

$$s_{tot} = \sum_j^{S \text{ or } N} s_{yj} , \quad (296)$$

$$S_{m0\alpha} = (\mathcal{S}_{m\alpha y} / s_{tot}) s_y , \quad (297)$$

$$S_{m1\alpha} = 0, \quad (298)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_\alpha = (\mathcal{S}_{m\alpha y}/s_{tot})s_y. \quad (299)$$

- WEST and EAST boundary:

$$s_{tot} = \sum_j^{W \text{ or } E} s_{xj}, \quad (300)$$

$$S_{m0\alpha} = (\mathcal{S}_{m\alpha x}/s_{tot})s_x, \quad (301)$$

$$S_{m1\alpha} = 0, \quad (302)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_\alpha = (\mathcal{S}_{m\alpha x}/s_{tot})s_x. \quad (303)$$

BCMOM=10 ... Prescribe the Value of the Decay Length for the Momentum (Only for the SOUTH and NORTH Boundary)

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv l_{u_\alpha} = u_\alpha / \partial_y u_\alpha$$

$$S_{m0\alpha} = A_\alpha m_p u_\alpha S_{p0\alpha}, \quad (304)$$

$$S_{m1\alpha} = -\frac{\eta_a^\alpha}{l_{u_\alpha}} s_y + A_\alpha m_p S_{p1\alpha} n_\alpha, \quad (305)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha}u_\alpha = A_\alpha m_p u_\alpha S_{p\alpha} - \frac{\eta_a^\alpha}{l_{u_\alpha}} u_\alpha s_y. \quad (306)$$

BCMOM=11 ... Prescribe the Value of the Decay Length for the Momentum (Only for the SOUTH Boundary, Probably Error in the Source Code!)

$$S_{m0\alpha} = \eta_a^\alpha n_\alpha A_\alpha \sqrt{1 - b_x^2} u_\alpha, \quad (307)$$

$$S_{m1\alpha} = -D_{an}^\alpha A_\alpha \frac{n_{\alpha(S)} - n_{\alpha(S+1)}}{h_{y1}} - \frac{\eta_a^\alpha}{h_{y1}} n_\alpha A_\alpha \sqrt{1 - b_x^2}, \quad (308)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha. \quad (309)$$

4.2.4 Boundary Conditions for Electron Energy Equation

BCENE=1 ... Prescribe the Value of the Electron Temperature

$$\text{ENEPAR}(\text{IB}, 1) \equiv \mathcal{T}_e$$

- SOUTH and NORTH boundary:

$$S_{he0} = \mathcal{V} \mathcal{N}_e \mathcal{T}_e e V s_y, \quad (310)$$

$$S_{he1} = -\mathcal{V} \mathcal{N}_e s_y, \quad (311)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = \mathcal{V} \mathcal{N}_e s_y \left(\mathcal{T}_e e V - \hat{T}_e \right). \quad (312)$$

- WEST and EAST boundary:

$$S_{he0} = \mathcal{U}\mathcal{N}_e\mathcal{T}_e eV s_x, \quad (313)$$

$$S_{he1} = -\mathcal{U}\mathcal{N}_e s_x, \quad (314)$$

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e = \mathcal{U}\mathcal{N}_e s_x \left(\mathcal{T}_e eV - \hat{T}_e \right). \quad (315)$$

BCENE=2 ... Prescribe the Value of the Gradient of the Electron Temperature

$$\text{ENEPAR}(\text{IB}, 1) \equiv \partial_x \mathcal{T}_e, \partial_y \mathcal{T}_e$$

- SOUTH boundary:

$$l_y = \frac{1}{2} \left(h_{y1(ix,S)} + h_{y1(ix,S+1)} \right), \quad (316)$$

$$S_{he0} = \mathcal{V}\mathcal{N}_e eV s_y \left[(-\partial_y \mathcal{T}_e) l_y + \hat{T}_e \right], \quad (317)$$

$$S_{he1} = -\mathcal{V}\mathcal{N}_e eV s_y, \quad (318)$$

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e = \mathcal{V}\mathcal{N}_e eV s_y \left[(-\partial_y \mathcal{T}_e) l_y \right]. \quad (319)$$

- NORTH boundary:

$$l_y = \frac{1}{2} (h_{y1(ix, N-1)} + h_{y1(ix, N)}) , \quad (320)$$

$$S_{he0} = \mathcal{V} \mathcal{N}_e e V s_y \left[(\partial_y \mathcal{T}_e) l_y + \widehat{T}_e \right] , \quad (321)$$

$$S_{he1} = -\mathcal{V} \mathcal{N}_e e V s_y , \quad (322)$$

$$S_{he} = S_{he0} + S_{he1} \widehat{T}_e = \mathcal{V} \mathcal{N}_e e V s_y [(\partial_y \mathcal{T}_e) l_y] . \quad (323)$$

- WEST boundary:

$$l_x = \frac{1}{2} (h_{x(W, iy)} + h_{x(W+1, iy)}) , \quad (324)$$

$$S_{he0} = \mathcal{U} \mathcal{N}_e e V s_x \left[(-\partial_x \mathcal{T}_e) l_x + \widehat{T}_e \right] , \quad (325)$$

$$S_{he1} = -\mathcal{U} \mathcal{N}_e e V s_x , \quad (326)$$

$$S_{he} = S_{he0} + S_{he1} \widehat{T}_e = \mathcal{U} \mathcal{N}_e e V s_x [(-\partial_x \mathcal{T}_e) l_x] . \quad (327)$$

- EAST boundary:

$$l_x = \frac{1}{2} \left(h_{x(E-1, iy)} + h_{x(E, iy)} \right), \quad (328)$$

$$S_{he0} = \mathcal{U} \mathcal{N}_e e V s_x \left[(\partial_x \mathcal{T}_e) l_x + \widehat{T}_e \right], \quad (329)$$

$$S_{he1} = -\mathcal{U} \mathcal{N}_e e V s_x, \quad (330)$$

$$S_{he} = S_{he0} + S_{he1} \widehat{T}_e = \mathcal{U} \mathcal{N}_e e V s_x [(\partial_x \mathcal{T}_e) l_x]. \quad (331)$$

BCENE=3 ... Sheath Conditions, Electron Energy Transmission Coefficient is the Free Input Parameter (Only for the WEST and EAST Boundary)

$$\text{ENEPAR}(\text{IB}, 1) \equiv \delta_e,$$

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv \mathcal{M}_{\alpha 1},$$

$$\text{MOMPAR}(1, \text{IB}, 2) \equiv \mathcal{M}_{12},$$

$$\text{POTPAR}(\text{IB}, 2) \equiv \Psi$$

- WEST boundary:

IF ($\mathcal{M}_{12} < 0.5$) THEN:

$$u_{out} = b_x \mathcal{M}_{11} c_s, \quad (332)$$

$$\Gamma_{eout} = \min(0, -u_{out} n_e - j_x / e_0), \quad (333)$$

ELSE ($\mathcal{M}_{12} \geq 0.5$):

$$u_{test\alpha} = u_{\alpha(W+1, iy)} \left(1 + \frac{h_{x(W+1, iy)}}{h_{x(W+2, iy)}} \right) - u_{\alpha(W+2, iy)} \frac{h_{x(W+1, iy)}}{h_{x(W+2, iy)}}, \quad (334)$$

$$u_{out\alpha} = \max(b_x \mathcal{M}_{\alpha 1} c_{s\alpha}, -u_{test\alpha}), \quad (335)$$

$$\Gamma_{ex} = - \sum_{\alpha} u_{out\alpha} Z_{\alpha} n_{\alpha}, \quad (336)$$

$$\Gamma_{eout} = \min(0, \Gamma_{ex} - j_x / e_0). \quad (337)$$

$$S_{he0} = 0, \quad (338)$$

$$S_{he1} = \left[\delta_e + \frac{e_0 (\Phi_{(W)} - \Psi)}{\hat{T}_e} \right] \Gamma_{eout} s_x, \quad (339)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = \left[\delta_e + \frac{e_0 (\Phi_{(W)} - \Psi)}{\hat{T}_e} \right] \Gamma_{eout} \hat{T}_e s_x, \quad (340)$$

where the free input parameter Ψ is the electric potential at the magnetic presheath edge, i.e., at the boundary between the collisional presheath and collisionless magnetic presheath.

- EAST boundary:

IF ($\mathcal{M}_{12} < 0.5$) THEN:

$$u_{out} = b_x \mathcal{M}_{11} c_s, \quad (341)$$

$$\Gamma_{eout} = \max(0, u_{out} n_e - j_x / e_0), \quad (342)$$

ELSE ($\mathcal{M}_{12} \geq 0.5$):

$$u_{test\alpha} = u_{\alpha(E-2, iy)} \left(1 + \frac{h_{x(E-1, iy)}}{h_{x(E-2, iy)}} \right) - u_{\alpha(E-3, iy)} \frac{h_{x(E-1, iy)}}{h_{x(E-2, iy)}}, \quad (343)$$

$$u_{out\alpha} = \max(b_x \mathcal{M}_{\alpha 1} c_{s\alpha}, u_{test\alpha}), \quad (344)$$

$$\Gamma_{ex} = \sum_{\alpha} u_{out\alpha} Z_{\alpha} n_{\alpha}, \quad (345)$$

$$\Gamma_{eout} = \max(0, \Gamma_{ex} - j_x / e_0). \quad (346)$$

$$S_{he0} = 0, \quad (347)$$

$$S_{he1} = - \left[\delta_e + \frac{e_0 (\Phi_{(E)} - \Psi)}{\hat{T}_e} \right] \Gamma_{eout} s_x, \quad (348)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = - \left[\delta_e + \frac{e_0 (\Phi_{(E)} - \Psi)}{\hat{T}_e} \right] \Gamma_{eout} \hat{T}_e s_x, \quad (349)$$

where the free input parameter Ψ is the electric potential at the magnetic presheath edge, i.e., at the boundary between the collisional presheath and collisionless magnetic presheath.

BCENE=4 ... Prescribe the Value of the Electron Temperature "Weakly",
Using as the Control Parameter Some Prescribed Electron Flux

$$\text{ENEPAR}(\text{IB}, 1) \equiv \mathcal{T}_e,$$

$$\text{ENEPAR}(\text{IB}, 2) \equiv \Gamma_{ex}, \Gamma_{ey}$$

- SOUTH and NORTH boundary:

$$S_{he0} = \Gamma_{ey} \mathcal{T}_e s_y, \quad (350)$$

$$S_{he1} = -(\Gamma_{ey}/eV) s_y, \quad (351)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = \Gamma_{ey} s_y \left(\mathcal{T}_e - \hat{T}_e/eV \right). \quad (352)$$

- WEST and EAST boundary:

$$S_{he0} = \Gamma_{ex} \mathcal{T}_e s_x, \quad (353)$$

$$S_{he1} = -(\Gamma_{ex}/eV) s_x, \quad (354)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = \Gamma_{ex} s_x \left(\mathcal{T}_e - \hat{T}_e/eV \right). \quad (355)$$

BCENE=5 ... Prescribe the Value of the Electron Energy Flux per Unit Area

$$\text{ENEPAR}(\text{IB}, 1) \equiv Q_{hex}, Q_{hey}$$

- SOUTH and NORTH boundary:

$$S_{he0} = Q_{hey}s_y, \quad (356)$$

$$S_{he1} = 0, \quad (357)$$

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e = Q_{hey}s_y. \quad (358)$$

- WEST and NORTH boundary:

$$S_{he0} = Q_{hex}s_x, \quad (359)$$

$$S_{he1} = 0, \quad (360)$$

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e = Q_{hex}s_x. \quad (361)$$

BCENE=6 ... Prescribe the Value of the Total Electron Energy Flux for a Constant Electron Temperature (Only for the NORTH and SOUTH Boundary)

$$\text{ENEPAR}(\text{IB}, 1) \equiv \mathcal{Q}$$

The electron temperature $\hat{T}_1$, which is calculated from the given total electron energy flux $\mathcal{Q}$ (the physics behind that calculation is unclear to the author of this report), is used in the calculation of the boundary electron heat source.

$$S_{he0} = \mathcal{V} \mathcal{N}_e \hat{T}_1 s_y, \quad (362)$$

$$S_{he1} = -\mathcal{V} \mathcal{N}_e s_y, \quad (363)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = \mathcal{V} \mathcal{N}_e s_y \left(\hat{T}_1 - \hat{T}_e \right). \quad (364)$$

BCENE=7 ... Prescribe the Electron Temperature as a Function of Other Plasma Parameters (Not Yet Available)

BCENE=8 ... Prescribe the Value of the Total Electron Energy Flux with a Constant Flux Density

$$\text{ENEPAR}(\text{IB}, 1) \equiv Q_{hex}, Q_{hey}$$

- SOUTH and NORTH boundary:

$$s_{tot} = \sum_j^{S \text{ or } N} s_{yj}, \quad (365)$$

$$S_{he0} = (Q_{hey}/s_{tot}) s_y, \quad (366)$$

$$S_{he1} = 0, \quad (367)$$

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e = (Q_{hey}/s_{tot}) s_y. \quad (368)$$

- WEST and NORTH boundary:

$$s_{tot} = \sum_j^{W \text{ or } E} s_{xj}, \quad (369)$$

$$S_{he0} = (Q_{hex}/s_{tot}) s_x, \quad (370)$$

$$S_{he1} = 0, \quad (371)$$

$$S_{he} = S_{he0} + S_{he1}\hat{T}_e = (Q_{hex}/s_{tot}) s_x. \quad (372)$$

BCENE=9 ... **Prescribe the Value of the Decay Length for the Electron Temperature (Only for the SOUTH and NORTH Boundary)**

$$\text{ENEPAR}(\text{IB}, 1) \equiv l_{T_e} = \hat{T}_e / \partial_y \hat{T}_e$$

$$S_{he0} = \sum_{\alpha} Z_{\alpha} \frac{5}{2} \hat{T}_e S_{p0\alpha}, \quad (373)$$

$$S_{he1} = -\frac{\kappa_a^e}{l_{T_e}} s_y + \sum_{\alpha} Z_{\alpha} \frac{5}{2} S_{p1\alpha} n_{\alpha}, \quad (374)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = -\frac{\kappa_a^e}{l_{T_e}} \hat{T}_e s_y + \sum_{\alpha} Z_{\alpha} \frac{5}{2} \hat{T}_e S_{p\alpha}. \quad (375)$$

BCENE=10 ... **Feedback Option for the Core (Only for the SOUTH Boundary)**

$$S_{he0} = C_{he0} s_y, \quad (376)$$

$$S_{he1} = 0, \quad (377)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = C_{he0} s_y. \quad (378)$$

4.2.5 Boundary Conditions for Ion Energy Equation

BCENI=1 ... Prescribe the Value of the Ion Temperature

$$\text{ENIPAR}(\text{IB}, 1) \equiv \mathcal{T}_i$$

- SOUTH and NORTH boundary:

$$S_{hi0} = \mathcal{V}\mathcal{N}_e \mathcal{T}_i e V s_y, \quad (379)$$

$$S_{hi1} = -\mathcal{V}\mathcal{N}_e s_y, \quad (380)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_e = \mathcal{V}\mathcal{N}_e s_y \left(\mathcal{T}_i e V - \hat{T}_i \right). \quad (381)$$

- WEST and EAST boundary:

$$S_{hi0} = \mathcal{U}\mathcal{N}_e \mathcal{T}_i e V s_x, \quad (382)$$

$$S_{hi1} = -\mathcal{U}\mathcal{N}_e s_x, \quad (383)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = \mathcal{U}\mathcal{N}_e s_x \left(\mathcal{T}_i e V - \hat{T}_i \right). \quad (384)$$

BCENI=2 ... Prescribe the Value of the Gradient of the Ion Temperature

$$\text{ENIPAR}(\text{IB}, 1) \equiv \partial_x \mathcal{T}_i, \partial_y \mathcal{T}_i$$

- SOUTH boundary:

$$l_y = \frac{1}{2} (h_{y1(ix,S)} + h_{y1(ix,S+1)}) , \quad (385)$$

$$S_{hi0} = \mathcal{V} \mathcal{N}_e e V s_y \left[(-\partial_y \mathcal{T}_i) l_y + \widehat{T}_i \right] , \quad (386)$$

$$S_{hi1} = -\mathcal{V} \mathcal{N}_e e V s_y , \quad (387)$$

$$S_{hi} = S_{hi0} + S_{hi1} \widehat{T}_i = \mathcal{V} \mathcal{N}_e e V s_y \left[(-\partial_y \mathcal{T}_i) l_y \right] . \quad (388)$$

- NORTH boundary:

$$l_y = \frac{1}{2} (h_{y1(ix,N-1)} + h_{y1(ix,N)}) , \quad (389)$$

$$S_{hi0} = \mathcal{V} \mathcal{N}_e e V s_y \left[(\partial_y \mathcal{T}_i) l_y + \widehat{T}_i \right] , \quad (390)$$

$$S_{hi1} = -\mathcal{V} \mathcal{N}_e e V s_y , \quad (391)$$

$$S_{hi} = S_{hi0} + S_{hi1} \widehat{T}_i = \mathcal{V} \mathcal{N}_e e V s_y \left[(\partial_y \mathcal{T}_i) l_y \right] . \quad (392)$$

- WEST boundary:

$$l_x = \frac{1}{2} (h_{x(W, iy)} + h_{x(W+1, iy)}) , \quad (393)$$

$$S_{hi0} = \mathcal{UN}_e e V s_x \left[(-\partial_x \mathcal{T}_i) l_x + \widehat{T}_i \right] , \quad (394)$$

$$S_{hi1} = -\mathcal{UN}_e e V s_x , \quad (395)$$

$$S_{hi} = S_{hi0} + S_{hi1} \widehat{T}_i = \mathcal{UN}_e e V s_x \left[(-\partial_x \mathcal{T}_i) l_x \right] . \quad (396)$$

- EAST boundary:

$$l_x = \frac{1}{2} (h_{x(E-1, iy)} + h_{x(E, iy)}) , \quad (397)$$

$$S_{hi0} = \mathcal{UN}_e e V s_x \left[(\partial_x \mathcal{T}_i) l_x + \widehat{T}_i \right] , \quad (398)$$

$$S_{hi1} = -\mathcal{UN}_e e V s_x , \quad (399)$$

$$S_{hi} = S_{hi0} + S_{hi1} \widehat{T}_i = \mathcal{UN}_e e V s_x \left[(\partial_x \mathcal{T}_i) l_x \right] . \quad (400)$$

BCENI=3 ... Sheath Conditions, Ion Energy Transmission Coefficients Are the Free Input Parameters (Only for the WEST and EAST Boundary)

$$\text{ENIPAR}(\text{IB}, 1) \equiv \delta_{i1},$$

$$\text{ENIPAR}(\text{IB}, 2) \equiv \delta_{i2},$$

$$\text{MOMPAR}(\text{IS}, \text{IB}, 1) \equiv \mathcal{M}_{\alpha 1},$$

$$\text{MOMPAR}(1, \text{IB}, 2) \equiv \mathcal{M}_{12},$$

$$\text{POTPAR}(\text{IB}, 2) \equiv \Psi$$

- WEST boundary:

IF ($\mathcal{M}_{12} < 0.5$) THEN:

$$u_{out} = b_x \mathcal{M}_{11} c_s, \quad (401)$$

$$u_p = \frac{u_{out}}{b_x} + \sqrt{\frac{1}{b_x^2} - 1} u_{D1}, \quad (402)$$

$$S_{hi0} = - \sum_{\alpha} \delta_{i2} \frac{1}{2} u_{out} A_{\alpha} m_p n_{\alpha} (u_p^2 + u_{D\alpha}^2) s_x, \quad (403)$$

$$S_{hi1} = - \sum_{\alpha} u_{out} \delta_{i1} n_{\alpha} s_x, \quad (404)$$

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i = - \sum_{\alpha} n_{\alpha} u_{out} s_x \left[\delta_{i2} \frac{1}{2} A_{\alpha} m_p (u_p^2 + u_{D\alpha}^2) + \delta_{i1} \hat{T}_i \right], \quad (405)$$

ELSE ($\mathcal{M}_{12} \geq 0.5$):

$$u_{test\alpha} = u_{\alpha(W+1, iy)} \left(1 + \frac{h_{x(W+1, iy)}}{h_{x(W+2, iy)}} \right) - u_{\alpha(W+2, iy)} \frac{h_{x(W+1, iy)}}{h_{x(W+2, iy)}}, \quad (406)$$

$$u_{out\alpha} = \max(b_x \mathcal{M}_{\alpha 1} c_{s\alpha}, -u_{test\alpha}), \quad (407)$$

$$u_{p\alpha} = \frac{u_{out\alpha}}{b_x} + \sqrt{\frac{1}{b_x^2} - 1} u_{D\alpha}, \quad (408)$$

$$S_{hi0} = - \sum_{\alpha} \delta_{i2} \frac{1}{2} u_{out\alpha} A_{\alpha} m_p n_{\alpha} (u_{p\alpha}^2 + u_{D\alpha}^2) s_x, \quad (409)$$

$$S_{hi1} = - \sum_{\alpha} u_{out\alpha} \delta_{i1} n_{\alpha} s_x, \quad (410)$$

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i = - \sum_{\alpha} n_{\alpha} u_{out\alpha} s_x \left[\delta_{i2} \frac{1}{2} A_{\alpha} m_p (u_{p\alpha}^2 + u_{D\alpha}^2) + \delta_{i1} \hat{T}_i \right]. \quad (411)$$

- EAST boundary:

IF ($\mathcal{M}_{12} < 0.5$) THEN:

$$u_{out} = b_x \mathcal{M}_{11} c_s, \quad (412)$$

$$u_p = \frac{u_{out}}{b_x} - \sqrt{\frac{1}{b_x^2} - 1} u_{D1}, \quad (413)$$

$$S_{hi0} = - \sum_{\alpha} \delta_{i2} \frac{1}{2} u_{out} A_{\alpha} m_p n_{\alpha} (u_p^2 + u_{D\alpha}^2) s_x, \quad (414)$$

$$S_{hi1} = - \sum_{\alpha} u_{out} \delta_{i1} n_{\alpha} s_x, \quad (415)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = - \sum_{\alpha} n_{\alpha} u_{out} s_x \left[\delta_{i2} \frac{1}{2} A_{\alpha} m_p (u_p^2 + u_{D\alpha}^2) + \delta_{i1} \hat{T}_i \right], \quad (416)$$

ELSE ($\mathcal{M}_{12} \geq 0.5$):

$$u_{test\alpha} = u_{\alpha(E-2, iy)} \left(1 + \frac{h_{x(E-1, iy)}}{h_{x(E-2, iy)}} \right) - u_{\alpha(E-3, iy)} \frac{h_{x(E-1, iy)}}{h_{x(E-2, iy)}}, \quad (417)$$

$$u_{out\alpha} = \max(b_x \mathcal{M}_{\alpha 1} c_{s\alpha}, u_{test\alpha}), \quad (418)$$

$$u_{p\alpha} = \frac{u_{out\alpha}}{b_x} - \sqrt{\frac{1}{b_x^2} - 1} u_{D\alpha}, \quad (419)$$

$$S_{hi0} = - \sum_{\alpha} \delta_{i2} \frac{1}{2} u_{out\alpha} A_{\alpha} m_p n_{\alpha} (u_{p\alpha}^2 + u_{D\alpha}^2) s_x, \quad (420)$$

$$S_{hi1} = - \sum_{\alpha} u_{out\alpha} \delta_{i1} n_{\alpha} s_x, \quad (421)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = - \sum_{\alpha} n_{\alpha} u_{out\alpha} s_x \left[\delta_{i2} \frac{1}{2} A_{\alpha} m_p (u_{p\alpha}^2 + u_{D\alpha}^2) + \delta_{i1} \hat{T}_i \right]. \quad (422)$$

BCENI=4 ... Prescribe the Value of the Ion Temperature "Weakly",
Using as the Control Parameter Some Prescribed Ion Flux

$$\text{ENIPAR}(\text{IB}, 1) \equiv \mathcal{T}_i,$$

$$\text{ENIPAR}(\text{IB}, 2) \equiv \Gamma_{ix}, \Gamma_{iy}$$

- SOUTH and NORTH boundary:

$$S_{hi0} = \Gamma_{iy} \mathcal{T}_i s_y, \quad (423)$$

$$S_{hi1} = -(\Gamma_{iy}/eV) s_y, \quad (424)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = \Gamma_{iy} s_y \left(\mathcal{T}_i - \hat{T}_i / eV \right). \quad (425)$$

- WEST and EAST boundary:

$$S_{hi0} = \Gamma_{ix} \mathcal{T}_i s_x, \quad (426)$$

$$S_{hi1} = -(\Gamma_{ix}/eV) s_x, \quad (427)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = \Gamma_{ix} s_x \left(\mathcal{T}_i - \hat{T}_i / eV \right). \quad (428)$$

BCENI=5 ... Prescribe the Value of the Ion Energy Flux per Unit Area

$$\text{ENIPAR}(\text{IB}, 1) \equiv Q_{hix}, Q_{hiy}$$

- SOUTH and NORTH boundary:

$$S_{hi0} = Q_{hiy}s_y, \quad (429)$$

$$S_{hi1} = 0, \quad (430)$$

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i = Q_{hiy}s_y. \quad (431)$$

- WEST and NORTH boundary:

$$S_{hi0} = Q_{hix}s_x, \quad (432)$$

$$S_{hi1} = 0, \quad (433)$$

$$S_{hi} = S_{hi0} + S_{hi1}\hat{T}_i = Q_{hix}s_x. \quad (434)$$

BCENI=6 ... Prescribe the Value of the Total Ion Energy Flux for a Constant Ion Temperature (Only for the NORTH and SOUTH Boundary)

$$\text{ENIPAR}(\text{IB}, 1) \equiv \mathcal{Q}$$

The ion temperature $\hat{T}_1$, which is calculated from the given total ion energy flux $\mathcal{Q}$ (the physics behind that calculation is unclear to the author of this report), is used in the calculation of the boundary ion heat source.

$$S_{hi0} = \mathcal{V} \mathcal{N}_e \hat{T}_1 s_y, \quad (435)$$

$$S_{hi1} = -\mathcal{V} \mathcal{N}_e s_y, \quad (436)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = \mathcal{V} \mathcal{N}_e s_y \left(\hat{T}_1 - \hat{T}_i \right). \quad (437)$$

BCENI=7 ... Prescribe the Ion Temperature as a Function of Other Plasma Parameters (Not Yet Available)

BCENI=8 ... Prescribe the Value of the Total Ion Energy Flux with a Constant Flux Density

$$\text{ENIPAR}(\text{IB}, 1) \equiv Q_{hix}, Q_{hiy}$$

- SOUTH and NORTH boundary:

$$s_{tot} = \sum_j^{S \text{ or } N} s_{yj}, \quad (438)$$

$$S_{hi0} = (Q_{hiy}/s_{tot}) s_y, \quad (439)$$

$$S_{hi1} = 0, \quad (440)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = (Q_{hiy}/s_{tot}) s_y. \quad (441)$$

- WEST and NORTH boundary:

$$s_{tot} = \sum_j^{W \text{ or } E} s_{xj}, \quad (442)$$

$$S_{hi0} = (Q_{hix}/s_{tot}) s_x, \quad (443)$$

$$S_{hi1} = 0, \quad (444)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = (Q_{hix}/s_{tot}) s_x. \quad (445)$$

BCENI=9 ... Prescribe the Value of the Decay Length for the Ion Temperature (Only for the SOUTH and NORTH Boundary)

$$\text{ENIPAR}(\text{IB}, 1) \equiv l_{T_i} = \hat{T}_i / \partial_y \hat{T}_i$$

$$S_{hi0} = \sum_{\alpha} Z_{\alpha} \frac{5}{2} \hat{T}_i S_{p0\alpha}, \quad (446)$$

$$S_{hi1} = -\frac{\kappa_a^i}{l_{T_i}} s_y + \sum_{\alpha} Z_{\alpha} \frac{5}{2} S_{p1\alpha} n_{\alpha}, \quad (447)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = -\frac{\kappa_a^i}{l_{T_i}} \hat{T}_i s_y + \sum_{\alpha} Z_{\alpha} \frac{5}{2} \hat{T}_i S_{p\alpha}. \quad (448)$$

BCENI=10 ... Feedback Option for the Core (Only for the SOUTH Boundary)

$$S_{hi0} = C_{hi0} s_y, \quad (449)$$

$$S_{hi1} = 0, \quad (450)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = C_{hi0} s_y. \quad (451)$$

4.2.6 Boundary Conditions for the Electric Potential Equation

BCPOT=1 ... Prescribe the Value of the Potential

$$\text{POTPAR}(\text{IB}, 1) \equiv \Psi$$

- SOUTH and NORTH boundary:

$$S_{ch0} = \mathcal{V}\Psi s_y, \quad (452)$$

$$S_{ch1} = -\mathcal{V}s_y, \quad (453)$$

$$S_{ch} = S_{ch0} + S_{ch1}\Phi = \mathcal{V}s_y(\Psi - \Phi). \quad (454)$$

- WEST and EAST boundary:

$$S_{ch0} = \mathcal{U}\Psi s_x, \quad (455)$$

$$S_{ch1} = -\mathcal{U}s_x, \quad (456)$$

$$S_{ch} = S_{ch0} + S_{ch1}\Phi = \mathcal{U}s_x(\Psi - \Phi), \quad (457)$$

where Ψ is the potential at the magnetic presheath edge and Φ is the potential at the boundary.

BCPOT=2 ... Prescribe the Value of the Gradient of the Potential

$$\text{POTPAR}(\text{IB}, 1) \equiv \partial_x \Psi, \partial_y \Psi$$

- SOUTH boundary:

$$l_y = \frac{1}{2} (h_{y1(ix,S)} + h_{y1(ix,S+1)}) , \quad (458)$$

$$S_{ch0} = \mathcal{V} s_y [(-\partial_y \Psi) l_y + \Phi] , \quad (459)$$

$$S_{ch1} = -\mathcal{V} s_y , \quad (460)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = \mathcal{V} s_y [(-\partial_y \Psi) l_y] . \quad (461)$$

- NORTH boundary:

$$l_y = \frac{1}{2} (h_{y1(ix,N-1)} + h_{y1(ix,N)}) , \quad (462)$$

$$S_{ch0} = \mathcal{V} s_y [(\partial_y \Psi) l_y + \Phi] , \quad (463)$$

$$S_{ch1} = -\mathcal{V} s_y , \quad (464)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = \mathcal{V} s_y [(\partial_y \Psi) l_y] . \quad (465)$$

- WEST boundary:

$$l_x = \frac{1}{2} \left(h_{x(W, iy)} + h_{x(W+1, iy)} \right), \quad (466)$$

$$S_{ch0} = \mathcal{U} s_x [(-\partial_x \Psi) l_x + \Phi], \quad (467)$$

$$S_{ch1} = -\mathcal{U} s_x, \quad (468)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = \mathcal{U} s_x [(-\partial_x \Psi) l_x]. \quad (469)$$

- EAST boundary:

$$l_x = \frac{1}{2} \left(h_{x(E-1, iy)} + h_{x(E, iy)} \right), \quad (470)$$

$$S_{ch0} = \mathcal{U} s_x [(\partial_x \Psi) l_x + \Phi], \quad (471)$$

$$S_{ch1} = -\mathcal{U} s_x, \quad (472)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = \mathcal{U} s_x [(\partial_x \Psi) l_x]. \quad (473)$$

BCPOT=3 ... Sheath Conditions (Only for the WEST and EAST Boundary)

$$\text{POTPAR}(\text{IB}, 2) \equiv \Psi$$

- WEST boundary:

$$j_i = -e_0 \sum_{\alpha} Z_{\alpha} \Gamma_{\alpha x}, \quad (474)$$

$$j_e = \frac{1}{\sqrt{2\pi}} e_0 n_e v_{th,e} S_1 \exp \left[-\frac{e_0 (\Phi - \Psi)}{\widehat{T}_e} \right], \quad (475)$$

$$S_{ch0} = (1 - \gamma_{se}) j_e - j_i + \max(j_i, (1 - \gamma_{se}) j_e) e_0 \Phi / \widehat{T}_e, \quad (476)$$

$$S_{ch1} = -\max(j_i, (1 - \gamma_{se}) j_e) e_0 / \widehat{T}_e, \quad (477)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = (1 - \gamma_{se}) j_e - j_i. \quad (478)$$

- EAST boundary:

$$j_i = e_0 \sum_{\alpha} Z_{\alpha} \Gamma_{\alpha x}, \quad (479)$$

$$j_e = \frac{1}{\sqrt{2\pi}} e_0 n_e v_{th,e} S_1 \exp \left[-\frac{e_0 (\Phi - \Psi)}{\widehat{T}_e} \right], \quad (480)$$

$$S_{ch0} = (1 - \gamma_{se}) j_e - j_i + \max(j_i, (1 - \gamma_{se}) j_e) e_0 \Phi / \widehat{T}_e, \quad (481)$$

$$S_{ch1} = -\max(j_i, (1 - \gamma_{se}) j_e) e_0 / \widehat{T}_e, \quad (482)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = (1 - \gamma_{se}) j_e - j_i. \quad (483)$$

BCPOT=4 ... Prescribe the Value of the Potential "Weakly", Using the Prescribed Control Parameter, Which Corresponds to the Specific Electric Conductivity

$$\text{POTPAR}(\text{IB}, 1) \equiv \Psi,$$

$$\text{POTPAR}(\text{IB}, 2) \equiv \sigma_x, \sigma_y$$

- SOUTH and NORTH boundary:

$$S_{ch0} = \sigma_y \Psi s_y, \quad (484)$$

$$S_{ch1} = -\sigma_y s_y, \quad (485)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = \sigma_y s_y (\Psi - \Phi). \quad (486)$$

- WEST and EAST boundary:

$$S_{ch0} = \sigma_x \Psi s_x, \quad (487)$$

$$S_{ch1} = -\sigma_x s_x, \quad (488)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = \sigma_x s_x (\Psi - \Phi). \quad (489)$$

BCPOT=5 ... Prescribe the Value of the Current Flux Density per Unit Area

$$\text{POTPAR}(\text{IB}, 1) \equiv j_x, j_y \quad (490)$$

- SOUTH and NORTH boundary:

$$S_{ch0} = j_y s_y, \quad (491)$$

$$S_{ch1} = 0, \quad (492)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = j_y s_y. \quad (493)$$

- WEST and EAST boundary:

$$S_{ch0} = j_x s_x, \quad (494)$$

$$S_{ch1} = 0, \quad (495)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = j_x s_x. \quad (496)$$

BCPOT=6 ... Prescribe the Value of the Total Current Flux Density for a Constant Potential (Not Yet Available)

BCPOT=7 ... Prescribe the Potential as a Function of Other Plasma Parameters (Not Yet Available)

BCPOT=8 ... Prescribe the Value of the Total Electric Current with a Constant Flux Density

$$\text{POTPAR}(\text{IB}, 1) \equiv J_x, J_y$$

- SOUTH and NORTH boundary:

$$s_{tot} = \sum_j^{S \text{ or } N} s_{yj}, \quad (497)$$

$$S_{ch0} = (J_y/s_{tot}) s_y, \quad (498)$$

$$S_{ch1} = 0, \quad (499)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = (J_y/s_{tot}) s_y. \quad (500)$$

- WEST and EAST boundary:

$$s_{tot} = \sum_j^{W \text{ or } E} s_{xj}, \quad (501)$$

$$S_{ch0} = (J_x/s_{tot}) s_x, \quad (502)$$

$$S_{ch1} = 0, \quad (503)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = (J_x/s_{tot}) s_x. \quad (504)$$

BCPOT=9 ... Prescribe the Value of the Decay Length for the Potential (Only for the SOUTH and NORTH Boundary)

$$\text{POTPAR}(\text{IB}, 1) \equiv l_\Phi = \Phi / \partial_y \Phi$$

$$S_{ch0} = 0, \quad (505)$$

$$S_{ch1} = -\frac{\sigma_a s_y}{l_\Phi e_0 n_e},^{(*)} \quad (506)$$

$$S_{ch} = S_{ch0} + S_{ch1} \Phi = -\frac{\sigma_a \Phi s_y}{l_\Phi e_0 n_e}.^{(*)} \quad (507)$$

^(*)WARNING: PROBABLY ERROR OR THE SPECIFIC ELECTRIC CONDUCTIVITY IS GIVEN BY $\frac{\sigma_a}{e_0 n_e}$!

4.2.7 Interior Boundary at the "Dead" Regions

Particle Sources

$$S_{p0\alpha} = \mathcal{V} n_{\alpha(is)}, \quad (508)$$

$$S_{p1\alpha} = -\mathcal{V}, \quad (509)$$

$$S_{p\alpha} = S_{p0\alpha} + S_{p1\alpha} n_\alpha = \mathcal{V} (n_{\alpha(is)} - n_\alpha). \quad (510)$$

Momentum Sources

$$S_{m0\alpha} = 0, \quad (511)$$

$$S_{m1\alpha} = -\mathcal{V}\mathcal{N}_e A_\alpha s_y, \quad (512)$$

$$S_{m\alpha} = S_{m0\alpha} + S_{m1\alpha} u_\alpha = -\mathcal{V}\mathcal{N}_e A_\alpha s_y u_\alpha. \quad (513)$$

Electron Heat Source

$$S_{he0} = \mathcal{V}n_e \hat{T}_{e(is)} eV, \quad (514)$$

$$S_{he1} = -\mathcal{V}n_e, \quad (515)$$

$$S_{he} = S_{he0} + S_{he1} \hat{T}_e = \mathcal{V}n_e \left(\hat{T}_{e(is)} eV - \hat{T}_e \right). \quad (516)$$

Ion Heat Source

$$S_{hi0} = \mathcal{V}n_e \hat{T}_{i(is)} eV, \quad (517)$$

$$S_{hi1} = -\mathcal{V}n_e, \quad (518)$$

$$S_{hi} = S_{hi0} + S_{hi1} \hat{T}_i = \mathcal{V}n_e \left(\hat{T}_{i(is)} eV - \hat{T}_i \right). \quad (519)$$

Electric Charge Source

$$S_{ch0} = \mathcal{V}\Phi_{(is)}, \quad (520)$$

$$S_{ch1} = -\mathcal{V}, \quad (521)$$

$$S_{ch} = S_{ch0} + S_{ch1}\Phi = \mathcal{V}(\Phi_{(is)} - \Phi). \quad (522)$$

5 Conclusion

The existing expressions for the boundary conditions in the B2.5 code cannot adequately reproduce even relatively simple physical models of the magnetized boundary plasma. They are based on some unphysical or insufficiently justified assumptions about the boundary values of the particle, momentum and energy fluxes as functions of the plasma parameters calculated by the bulk transport equations. In order to improve the reliability and accuracy of the B2.5 code, following tasks related to the boundary conditions should be accomplished:

- realistic kinetic or improved fluid models, which take into account all relevant processes in the boundary plasma, such as solid surface processes, plasma particle collisions, electric and diamagnetic drifts etc., should be developed;
- analytical expressions for the boundary conditions, which can be directly applied into the B2.5 code, should be derived from these improved models of the boundary plasma;
- expressions for the boundary sources should be properly changed, so that these new analytical expressions for the boundary conditions can be implemented;
- expressions for the boundary sources should be modified, so that according to the theoretical models some more complicated non-linear dependences of the boundary conditions on the plasma parameters, can be implemented.

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